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of A as its columns and the columns of A as its rows.

For example, if $A = \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$

0 1 2 then $A^t = \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$

4 1

5 2 $\begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$. Thus, the transpose of a row vector is a column vector and vice-versa.

Theorem 1.2.2 For any matrix A , we have $(A^t)^t = A$. Proof. Let $A = [a_{ij}]$, $A^t = [b_{ij}]$ and $(A^t)^t = [c_{ij}]$. Then, the definition of transpose gives $c_{ij} =$

$b_{ji} = a_{ij}$ for all i, j and the result follows. Definition 1.2.3 (Addition of Matrices) Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $m \times n$ matrices. Then the sum $A + B$ is defined to be the matrix $C = [c_{ij}]$ with $c_{ij} = a_{ij} + b_{ij}$. Note that, we

define the sum of two matrices only when the order of the two matrices are same. Definition 1.2.4 (Multiplying a Scalar to a Matrix) Let $A = [a_{ij}]$ be an $m \times n$ matrix. Then for any element $k \in \mathbb{R}$, we define $kA = [ka_{ij}]$. For

example, if $A = \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$

0 1 2 and $k = 5$, then $5A = \begin{bmatrix} 5 & 20 & 25 \end{bmatrix}$

0 5 10 Theorem 1.2.5 Let A, B and C be matrices of order $m \times n$, and let $k, H \in \mathbb{R}$. Then 2. $(A + B) + C = A + (B + C)$ (associativity).

3. $k(HA) = (kH)A$.

Proof. Part 1.

Let $A = [a_{ij}]$ and $B = [b_{ij}]$. Then

$A + B = [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}] = [b_{ij} + a_{ij}] = [b_{ij}] + [a_{ij}] = B + A$ as real numbers commute. The reader is required to prove the other parts as all the results follow from the properties of real numbers. Exercise 1.2.6 1.

Suppose $A + B = A$. Then show that $B = 0$. 2. Suppose $A + B = 0$. Then show that $B = (-1)A = [-a_{ij}]$. Definition 1.2.7 (Additive Inverse) Let A be an $m \times n$ matrix. 1. Then there exists a matrix B with $A + B = 0$. This matrix

B is called the additive inverse of A , and is denoted by $-A = (-1)A$.

1. $A + B = B + A$ (commutativity).

2. Also, for the matrix $0_{m \times n}$, $A + 0 = 0 + A = A$. Hence, the matrix $0_{m \times n}$ is called the additive identity.

4. $(k + H)A = kA + HA$.

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Part 5. For all $j = 1, 2, \dots, n$, we have

$= 0$ whenever $i \neq k$. Hence, the required result follows. The reader is required to prove the other parts. Exercise 1.2.12 1. Let A and B be two matrices. If the matrix addition $A + B$ is defined, then prove

1.2.1 Multiplication of Matrices

Definition 1.2.8 (Matrix Multiplication / Product) Let $A = [a_{ij}]$ be an $m \times n$ matrix and $B = [b_{ij}]$ be an $n \times r$ matrix. The product AB is a matrix $C = [c_{ij}]$ of order $m \times r$, with $a_{ik}b_{kj} = a_{11}b_{1j} + a_{21}b_{2j} + \dots + a_{n1}b_{nj}$. Observe that the

product AB is defined if and only if the number of columns of A = the number of rows of B . For example, if $A = \begin{bmatrix} 1 & 2 & 3 & 2 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$

0 0 3

1 0 4 $\begin{bmatrix} 1 & 2 & 3 & 2 & 4 & 1 \end{bmatrix}$ then

$AB = 1 + 0 + 3 + 2 + 0 + 0 + 1 + 6 + 12$

$2 + 0 + 1 + 4 + 0 + 0 + 2 + 12 + 4 = 4 + 2 + 19$

3 4 18 Note that in this example, while AB is defined, the product BA is not defined. However, for square matrices A and B of the same order, both the product AB and BA are defined. Definition 1.2.9 Two square

matrices A and B are said to commute if $AB = BA$. Remark 1.2.10 1. Note that if A is a square matrix of order n then $AI_n = I_n A = A$. Also for any $d \in \mathbb{R}$, the matrix dI_n commutes with every square matrix of order n . The matrices

for any $d \in \mathbb{R}$ are called scalar matrices. 2. In general, the matrix product is not commutative. For example, consider the following two matrices $A = \begin{bmatrix} 1 & 1 \end{bmatrix}$

0 0 and $B = \begin{bmatrix} 1 & 0 \end{bmatrix}$

1 0. Then check that the matrix product

$0_{ij} = a^{ij}$

$AB = 2 \ 0$

0 0 $\approx 1 \ 1$

$k=1$

1 1 = BA .

Theorem 1.2.11 Suppose that the matrices A, B and C are so chosen that the matrix multiplications are defined. 1. Then $(AB)C = A(BC)$. That is, the matrix multiplication is associative.

2. For any $k \in \mathbb{R}$, $(kA)B = k(AB) = A(kB)$.

3. Then $A(B + C) = AB + AC$. That is, multiplication distributes over addition.

4. If A is an $n \times n$ matrix then $AI_n = I_n A = A$. 5. For any square matrix A of order n and $D = \text{diag}(d_1, d_2, \dots, d_n)$, we have the first row of DA is

d_1 times the first row of A ;

for $1 \leq i \leq n$, the i th row of DA is d_i times the i th row of A . A similar statement holds for the columns of A when A is multiplied on the right by D . Proof. Part 1. Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{n \times p}$ and $C = [c_{ij}]_{p \times q}$. Then $b_{1p}c_{1q}$ and

$(AB)_{1p} = a^{1n}$

14 CHAPTER 1. MATRICES 2. Let $A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 3 & 3 & 3 & 3 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$. Then A is an orthogonal matrix.

3. Let $A = [a_{ij}]$ be an $n \times n$ matrix with $a_{ij} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 3 & 3 & 3 & 3 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$ if $i = j + 1$ and $n - 1$. The matrices A for which a positive integer k exists such that $A^k = 0$ are called nilpotent matrices. The least positive

integer k for which $A^k = 0$ is called the order of nilpotency. 4. Let $A = \begin{bmatrix} 1 & 0 \end{bmatrix}$

0 0. Then $A^2 = A$. The matrices that satisfy the condition that $A^2 = A$ are called

$(BC)_{ij} = a^{ip}$

$a_{ik}b_{kj}$

$d=1$

$k=1$

a^{ip}

a^{ip}

idempotent matrices. Exercise 1.3.3 1. Show that for any square matrix A , $S = \frac{1}{2}(A + A^t)$ is symmetric, $T = \frac{1}{2}(A - A^t)$ is skew-symmetric, and $A = S + T$. 2. Show that the product of two lower triangular matrices is a lower

triangular matrix. A similar statement holds for upper triangular matrices. 3. Let A and B be symmetric matrices. Show that AB is symmetric if and only if $AB = BA$.

that $(A + B)^t = A^t + B^t$. Also, if the matrix product AB is defined then prove that $(AB)^t = B^t A^t$.

4. Show that the diagonal entries of a skew-symmetric matrix are zero. 5. Let A, B be skew-symmetric matrices with $AB = BA$. Is the matrix AB symmetric or skew-symmetric? 6. Let A be a symmetric matrix of order n with $A^2 = 0$. Is it necessarily true that $A = 0$?

2. Let $A = [a_1, a_2, \dots, a_n]$ and $B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 3 & 3 & 3 & 3 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$

Therefore, $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 3 & 3 & 3 & 3 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$. Compute the matrix products AB and BA . $a_{ik} = a^{ip}$

$A(BC)_{ij} = a^{ip}$

$a_{ik} BC_{kj} = a^{ip}$

$b_{1p}c_{1q}$

3. Let n be a positive integer. Compute A^n for the following matrices:

$k=1$

$k=1$

$d=1$

a^{ip}

a^{ip}

$= a^{ip}$

$$a_{ik}b_{k0}c_{0j} = \sum_{k=1}^n a_{ik}b_{k0}c_{0j}$$

$$0 \ 1 \ , \ \begin{bmatrix} & & 1 & 1 & 1 \\ & & & & \\ 0 & 0 & 1 & & \\ & & & & \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ & & \\ & & \end{bmatrix}$$

$$a_{ik}b_{k0}c_{0j}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= \sum_{k=1}^n$$

7. Let A be a nilpotent matrix. Show that there exists a matrix B such that $B(I + A) = I = (I + A)B$.

$$\sqrt[3]{6} \sqrt[3]{6} - \sqrt[3]{6}$$

Definition 1.3.4 A matrix obtained by deleting some of the rows and/or columns of a matrix is said to be a submatrix of

1.3.1 Submatrix of a Matrix

the given matrix.

For example, if $A = \begin{bmatrix} 1 & 4 & 5 & 0 \end{bmatrix}$ otherwise. Then $A^0 = 0$ and $A^0 = 0$ for $1 \leq h \leq$

$$a_{ik}b_{k0}c_{0j} = \sum_{k=1}^n a_{ik}b_{k0}c_{0j}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ & & \\ & & \end{bmatrix}$$

Can you guess a formula for A^n and prove it by induction?

$$AB = \sum_{k=1}^n a_{ik}b_{k0}c_{0j}$$

4. Find examples for the following statements. (a) Suppose that the matrix product AB is defined. Then the product BA need not be defined. $(DA)_j = \sum_{k=1}^n$

$$0 \ 1 \ 2 \ , \text{ a few submatrices of } A \text{ are } [1], [2], \begin{bmatrix} 1 \\ 0 \end{bmatrix}, [1 \ 5], \begin{bmatrix} 1 & 5 \end{bmatrix}$$

0 2 . But the matrices 1 4
(b) Suppose that the matrix products AB and BA are defined. Then the matrices AB and BA can have different orders. (c) Suppose that the matrices A and B are square matrices of order n . Then AB and BA may or may not be equal. $\sum_{k=1}^n$

$$1 \ 0 \text{ and } 1 \ 4$$

are not submatrices of A . (The reader is advised to give reasons.)

Miscellaneous Exercises

Exercise 1.3.5 1. Complete the proofs of Theorems 1.2.5 and 1.2.11.

$$2. \text{ Let } x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

1.3 Some More Special Matrices

Definition 1.3.1 1. A matrix A over R is called symmetric if $A^t = A$ and skew-symmetric if $A^t = -A$.

$$\begin{bmatrix} & \\ & \end{bmatrix}$$

2. A matrix A is said to be orthogonal if $AA^t = A^tA = I$. Example 1.3.2 1. Let $A = \begin{bmatrix} & & 1 & 2 & 3 \\ & & & & \\ 2 & & & & \end{bmatrix}$.

$$\begin{bmatrix} 3 & -1 & 4 \\ & & \end{bmatrix} \text{ and } B = \begin{bmatrix} & 0 & 1 & 2 & -1 & 0 & -3 \\ & & & & & & \\ -2 & 3 & 0 & & & & \end{bmatrix}$$

Then A is a symmetric matrix and B is a skew-symmetric matrix.

$A = \begin{bmatrix} 1 & 0 \\ & \end{bmatrix}$
Geometrically interpret $y = Ax$
3. Consider the two coordinate transformations

$$\text{and } B = \begin{bmatrix} \cos \theta & -\sin \theta \\ 0 & -1 \end{bmatrix}$$

$$\sin \theta \cos \theta$$

$$y_2$$

$$\text{and } y = Bx.$$

$$x_1 = a_{11}y_1 + a_{12}y_2$$

$$x_2 = a_{21}y_1 + a_{22}y_2 \text{ and } y_1 = b_{11}z_1 + b_{12}z_2, y_2 = b_{21}z_1 + b_{22}z_2.$$

1.3. SOME MORE SPECIAL MATRICES 15 (a) Compose the two transformations to express x_1, x_2 in terms of z_1, z_2 .

(b) If $x^t = [x_1, x_2]$, $y^t = [y_1, y_2]$ and $z^t = [z_1, z_2]$ then find matrices A, B and C such that (c) Is $C = AB$?

4. For a square matrix A of order n , we define trace of A , denoted by $\text{tr}(A)$ as

$$\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn} \quad \text{(a) } \text{tr}(A + B) = \text{tr}(A) + \text{tr}(B). \quad \text{(b) } \text{tr}(AB) = \text{tr}(BA).$$

5. Show that, there do not exist matrices A and B such that $AB - BA = cI_n$ for any $c \neq 0$.

16 CHAPTER 1. MATRICES Theorem 1.3.6 is very useful due to the following reasons: 1. The order of the matrices P, Q, H and K are smaller than that of A or B . 2. It may be possible to block the matrix in such a way

that a few blocks are either identity matrices or zero matrices. In this case, it may be easy to handle the matrix product using the block form: 3. Or when we want to prove results using induction, then we may assume the

result for $r \times r$ submatrices and then look for $(r + 1) \times (r + 1)$ submatrices, etc.

$$\text{For example, if } A = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} & & a & b \\ & & & \\ & & & c & d \end{bmatrix}$$

$$\text{e f } \begin{bmatrix} & \\ & \end{bmatrix}, \text{ Then } \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} 0 & \\ & \end{bmatrix} [e \ f] = \begin{bmatrix} a + 2c & b + 2d \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ & \end{bmatrix}$$

6. Let A and B be two $m \times n$ matrices and let x be an $n \times 1$ column vector. (a) Prove that if $Ax = 0$ for all x , then A is the zero matrix. (b) Prove that if $Ax = Bx$ for all x , then $A = B$.

$$\begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} 0 & -1 & 2 \end{bmatrix}$$

$$2a + 5c \ 2b + 5d$$

7. Let A be an $n \times n$ matrix such that $AB = BA$ for all $n \times n$ matrices B . Show that $A = cI$ for some c . Let $A = \begin{bmatrix} & & 1 & 2 \\ & & & \\ 3 & 1 & & \end{bmatrix}$

. Show that there exist infinitely many matrices B such that $BA = I_2$. Also, show $2 \ 1$

that there does not exist any matrix C such that $AC = I_3$.
 $x = Ay, y = Bz$ and $x = Cz$.

1.3.1 Block Matrices

$-2 \ 5 \ -3$, then A can be decomposed as follows: $3 \ 1 \ 4$

Let A be an $n \times m$ matrix and B be an $m \times p$ matrix. Suppose $r < m$. Then, we can decompose the
Then the matrices P, Q, R, S and E, F, G, H , are called the blocks of the matrices A and B , respectively. Even if $A + B$ is defined,
G H

the orders of P and E may not be same and hence, we may not be able
to add A and B in the block form. But, if $A + B$ and $P + E$ is
defined then $A + B = \begin{bmatrix} P + E & Q + F \\ R + G & S + H \end{bmatrix}$. Similarly, if the product AB is defined, the product $P E$ need not be defined. Therefore, we

can talk of matrix product AB as block product of matrices, if both the products AB and P are defined. And in this case, we have $AB = P E + Q G P F + Q H$

matrices A and B as $A = \begin{bmatrix} P & Q \end{bmatrix}$ and $B = \begin{bmatrix} E & F \\ G & H \end{bmatrix}$; where P has order $n \times r$ and H has order $r \times p$. That is, the matrices P and Q are submatrices of A and P consists of the first r columns of A and Q consists of the last $m - r$ columns of A . Similarly, H and K are submatrices of B and H consists of the first r

R and S consists of the last $m - r$ rows of B . We now prove the following important theorem. Theorem 1.3.6 Let A

Proof. First note that the matrices $P H$ and $Q K$ are each of order $n \times p$. The matrix products $P H$ and $Q K$ are valid as the order of the matrices P, H, Q and K are respectively, $n \times r, r \times p, n \times (m - r)$ and $(m - r) \times p$. Let $P = [P_{ij}]$, $Q = [Q_{ij}]$, $H = [H_{ij}]$, and $K = [K_{ij}]$. Then, for $1 \leq i \leq n$ and $1 \leq j \leq p$, we have

$$\alpha \in \mathbb{R}.$$

$$\begin{bmatrix} 0 & -1 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}, \text{ or } A$$

$$= (P H)_j + (Q K)_j = (P H + Q K)_j.$$

$$\begin{bmatrix} 1 & 1 & 2 & 1 \end{bmatrix}$$

$$=$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

$R E + S G R F + S H$. That is, once a partition of A is fixed, the partition of B has to be properly chosen for purposes of block addition or multiplication. Exercise 1.3.7 1. Compute the matrix product AB using the block matrix multiplication for the matrices:

$$\text{If } A =$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 2 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} -2 & 5 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 2 \end{bmatrix}$$

$$2. \text{ Let } A = \begin{bmatrix} P & Q \\ R & S \end{bmatrix}.$$

If P, Q, R and S are symmetric, what can you say about A ? Are P, Q, R and S symmetric, when A is

$$\text{symmetric? Suppose } A = \begin{bmatrix} m_1 & m_2 \\ m_1 & n_1 \end{bmatrix}$$

$$r_2$$

$$r_1$$

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$$n_2$$

3. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices. Suppose $a_1, a_2, \dots, a_n$ are the rows of A and $b_1, b_2, \dots,$

b_p are the columns of B . If the product AB is defined, then show that

$$AB = [Ab_1, Ab_2, \dots, Ab_p] = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} a_1 B$$

$$(AB)_j = \sum_{k=1}^n a_{kj} b_{kj}$$

$$a_{kj} b_{kj} = \sum_{k=1}^n$$

$$a_{kj} b_{kj} + \sum_{k=1}^n$$

$$a_{kj} b_{kj}$$

[That is, left multiplication by A , is same as multiplying each column of B by A . Similarly, right multiplication by B , is same as multiplying each row of A by B .]

$$\sum_{k=1}^n$$

$$k=1$$

$$k=1$$

$$k=1$$

$$= \sum_{k=1}^n P_{kj} H_{kj} + \sum_{k=1}^n$$

$$Q_{kj} K_{kj}$$

$$k=1$$

1.4 Matrices over Complex Numbers

Here the entries of the matrix are complex numbers. All the definitions still hold. One just needs to look at the following additional definitions.

Definition 1.4.1 (Conjugate Transpose of a Matrix) 1. Let A be an $m \times n$ matrix

over C . If $A = [a_{ij}]$ then the Conjugate of A , denoted by A^* , is the matrix $B = [b_{ij}]$ with $b_{ij} = \overline{a_{ji}}$. For example, Let $A = \begin{bmatrix} 1 & 4 + 3i \\ 0 & 1 - i \end{bmatrix}$. Then $A^* = \begin{bmatrix} 1 & 4 - 3i \\ 0 & 1 + i \end{bmatrix}$.

2. Let A be an $m \times n$ matrix over C . If $A = [a_{ij}]$ then the Conjugate Transpose of A , denoted by A^* , is

the matrix $B = [b_{ij}]$ with $b_{ij} = \overline{a_{ji}}$. For example, Let $A = \begin{bmatrix} 1 & 4 + 3i \\ 0 & 1 - i \end{bmatrix}$. Then

$$\begin{bmatrix} 1 & 4 - 3i \\ 0 & 1 + i \end{bmatrix}$$

$$B^*.$$

$$a_2 B$$

$$a_2 B$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix}$$

3. A square matrix A over C is called Hermitian if $A^* = A$.

4. A square matrix A over C is called skew-Hermitian if $A^* = -A$. 5. A square matrix A over C is called unitary if $A^* A = A A^* = I$. 6. A square matrix A over C is called Normal if $AA^* = A^* A$. Remark 1.4.2 If $A = [a_{ij}]$ with $a_{ij} \in$

$\mathbb{R}$, then $A^* = A^T$. Exercise 1.4.3 1. Give examples of Hermitian, skew-Hermitian and unitary matrices that have entries with non-zero imaginary parts. 2. Restate the results on transpose in terms of conjugate transpose. 3.

Show that for any square matrix A , $S = A^* A$.

A is Hermitian, $T = A^* A$.

A is skew-Hermitian, and

4. Show that if A is a complex triangular matrix and $AA^* = A^* A$ then A is a diagonal matrix.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^* =$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -3i \\ -i & -2 \end{bmatrix}$$

$A = S + T$
 $A =$

Chapter 2

Linear System of Equations

2.1 Introduction

Let us look at some examples of linear systems.

- Suppose $a, b \in \mathbb{R}$. Consider the system $ax = b$. (a) If $a \neq 0$ then the system has a unique solution $x = \frac{b}{a}$.

(b) If $a = 0$ and $b \neq 0$ then the system has no solution.

ii. $b = 0$ then the system has infinite number of solutions, namely all $x \in \mathbb{R}$.

- We now consider a system with 2 equations in 2 unknowns. Consider the equation $ax + by = c$. If one of the coefficients, a or b is non-zero,

then this linear equation represents a line in $\mathbb{R}^2$. Thus for the system $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, the set of solutions is given by the points of intersection of the two lines. There are three cases to be considered.

Each case is illustrated by an example. (a) Unique Solution

$x + 2y = 1$ and $x + 3y = 1$. The unique solution is $(x, y)^t = (1, 0)^t$. Observe that in this case, $a_1b_2 - a_2b_1 \neq 0$.

(b) Infinite Number of Solutions

20 CHAPTER 2. LINEAR SYSTEM OF EQUATIONS (a) Unique Solution Consider the system $x+y+z = 3$, $x+4y+2z = 7$ and $4x+10y-z = 13$. The unique solution to this system is $(x, y, z)^t = (1, 1, 1)^t$; i.e. the three planes intersect at a point.

(b) Infinite Number of Solutions Consider the system $x + y + z = 3$, $x + 2y + 2z = 5$ and $3x + 4y + 4z = 11$. The set of solutions to this system is $(x, y, z)^t = (1, 2 - z, z)^t = (1, 2, 0)^t + z(0, -1, 1)^t$ with arbitrary z . The three planes intersect on a line.

(c) No Solution The system $x + y + z = 3$, $x + 2y + 2z = 5$ and $3x + 4y + 4z = 13$ has no solution. In this case, we get three parallel lines as intersections of the above planes taken two at a time.

The readers are advised to supply the proof.

2.2 Definition and a Solution Method

Definition 2.2.1 (Linear System) A linear system of m equations in n unknowns $x_1, x_2, \dots, x_n$ is a set

of equations of the form $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$ where for $1 \leq i \leq n$, and $1 \leq j \leq m$; $a_{ij}, b_i \in \mathbb{R}$. Linear System (2.2.1) is called homogeneous if $b_1 = 0 = b_2 = \dots = b_m$ and non-homogeneous otherwise. We rewrite the above equations in

the form $Ax = b$, where $a_{11} \ a_{12} \ \dots \ a_{1n}$

$$a_{21} \ a_{22} \ \dots \ a_{2n}$$

$$\begin{bmatrix} a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

The matrix A is called the coefficient matrix and the block matrix $[Ab]$, is the augmented matrix of the linear system (2.2.1).

Remark 2.2.2 Observe that the i th row of the augmented matrix $[A \ b]$ represents the i th equation and the j th column of the coefficient matrix A corresponds to coefficients of the j th variable x_j . That is, for $1 \leq i \leq m$ and $1 \leq j \leq n$, the entry a_{ij} of the coefficient matrix A corresponds to the i th equation

$x + 2y = 1$ and $2x + 4y = 2$. The set of solutions is $(x, y)^t = (1 - 2y, y)^t = (1, 0)^t + y(-2, 1)^t$ with y arbitrary. In other words, both the equations

represent the same line. Observe that in this case, $a_1b_2 - a_2b_1 = 0$, $a_1c_2 - a_2c_1 = 0$ and $b_1c_2 - b_2c_1 = 0$.

(c) No Solution

$x + 2y = 1$ and $2x + 4y = 3$. The equations represent a pair of parallel lines and hence there is no point of intersection.

th variable x_j . For a system of linear equations $Ax = b$, the system $Ax = 0$ is called the associated homogeneous system.

Definition 2.2.3 (Solution of a Linear System) A solution of the linear system $Ax = b$ is a column vector

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad (2.2.1)$$

y with entries $y_1, y_2, \dots, y_n$ such that the linear system (2.2.1) is satisfied by substituting y in place of x . That is, if $y^t = [y_1, y_2,$

$\dots, y_n]$ then $Ay = b$ holds. Note: The zero n -tuple $x = 0$ is always a solution of the system $Ax = 0$, and is called the trivial solution. A non-zero n -tuple x , if it satisfies $Ax = 0$, is called a non-trivial solution.

$$A =$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$b_m$$

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b².

2.2.1 A Solution Method Example 2.2.4 Let us solve the linear system $x + 7y + 3z = 11$, $x + y + z = 3$, and $4x + 10y - z = 13$.

the equation by $c \neq 0$; (compare the system (2.2.5) and the system (2.2.4).)

2. multiply a non-zero constant throughout an equation, say "multiply the k

1. The above linear system and the linear system

$x + y + z = 3$ Interchange the first two equations.

$x + 7y + 3z = 11$ (2.2.2)

$4x + 10y - z = 13$ have the same set of solutions. (why?)

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$x + y + z = 3$

$6y + 2z = 8$ (obtained by subtracting the first

equation from the second equation.)

$6y - 5z = 1$ (obtained by subtracting 4 times the first equation

from the third equation.) (2.2.3)

This system and the system (2.2.2) has the same set of solution. (why?)

3. Eliminating y from the last two equations of system (2.2.3), we get the system $x + y + z = 3$

Solution:

$6y + 2z = 8$

$7z = 7$ obtained by subtracting the third equation

from the second equation. (2.2.4)

which has the same set of solution as the system (2.2.3). (why?)

4. The system (2.2.4) and system

2.3. ROW OPERATIONS AND EQUIVALENT SYSTEMS 25 5. Multiply the 3rd equation by -2 (or $R_3(-2)$).

$$\begin{array}{ccc} z = 1 & \square & \square \\ y + z = 2 & \square & \square \\ 0 \ 1 \ 2 & \square & \square \end{array}$$

$0 \ 0 \ 1 \ 1$ $\square \square$ The last equation gives $z = 1$, the second equation now gives $y = 1$. Finally the first equation gives $x = 1$. Hence the set of solutions is $(x, y, z)^t = (1, 1, 1)^t$, a unique solution.

Example 2.3.11 Solve the linear system by Gauss elimination method.
 $R_1(1/2)$

$$\begin{array}{ccc} x + y + z = 3 \\ x + 2y + 2z = 5 \\ 3x + 4y + 4z = 11 \end{array}$$

Solution: In this case, the augmented matrix is $\square \square \square \begin{smallmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \\ 3 & 4 & 4 & 11 \end{smallmatrix}$ and the method proceeds as follows:

1. Add -1 times the first equation to the second equation. $1 \ 2 \ 2 \ 5$

$$\begin{array}{ccc} 3x + 4y + 4z = 11 & \square & \square \\ x + y + z = 3 & \square & \square \\ y + z = 2 & \square & \square \\ 0 \ 1 \ 2 & \square & \square \end{array}$$

$$3 \ 4 \ 4 \ 11 \square \square \square$$

2. Add -3 times the first equation to the third equation.

$$x + y + z = 3$$

$$\begin{array}{ccc} y + z = 2 & \square & \square \\ x + 3z = 5 & \square & \square \\ y + z = 2 & \square & \square \\ 0 \ 1 \ 2 & \square & \square \end{array}$$

$$0 \ 1 \ 2 \square \square$$

3. Add -1 times the second equation to the third equation

$$\begin{array}{ccc} y + z = 2 & \square & \square \\ x + y + z = 3 & \square & \square \\ 0 \ 1 \ 2 & \square & \square \end{array}$$

$0 \ 0 \ 0 \square \square$ Thus, the set of solutions is $(x, y, z)^t = (1, 2 - z, z)^t = (1, 2, 0)^t + z(0, -1, 1)^t$, with z arbitrary. In other words, the system has infinite number of solutions.

Example 2.3.12 Solve the linear system by Gauss

elimination method. $x + y + z = 3$

$$\begin{array}{ccc} x + 2y + 2z = 5 \\ 3x + 4y + 4z = 12 \end{array}$$

Solution: In this case, the augmented matrix is $\square \square \square \begin{smallmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \\ 3 & 4 & 4 & 12 \end{smallmatrix}$ and the method proceeds as follows:

1. Add -1 times the first equation to the second equation. $1 \ 2 \ 2 \ 5$

$$\begin{array}{ccc} 3x + 4y + 4z = 12 & \square & \square \\ x + y + z = 3 & \square & \square \\ y + z = 2 & \square & \square \\ 0 \ 1 \ 2 & \square & \square \end{array}$$

$$3 \ 4 \ 4 \ 12 \square \square \square$$

$$0 \ 0 \ -3 \ -2 \square \square \square$$

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$$x + 3z = 5$$

2. Add -3 times the first equation to the third equation.

$$x + y + z = 3$$

$$\begin{array}{ccc} y + z = 3 & \square & \square \\ y + z = 2 & \square & \square \\ 0 \ 1 \ 2 & \square & \square \end{array}$$

$$0 \ 1 \ 2 \square \square$$

$$0 \ 1 \ 3 \square \square$$

$$x + 3z = 5$$

3. Add -1 times the second equation to the third equation

$$y - \frac{1}{2}z = \frac{1}{2}$$

$$x + y + z = 3$$

$$\begin{array}{ccc} 0 = 1 & \square & \square \\ y + z = 2 & \square & \square \\ 0 \ 1 \ 2 & \square & \square \end{array}$$

$$0 \ 1 \ 2 \square \square$$

$$x + 3z = 5$$

$0 \ 0 \ 1 \square \square \square$ The third equation in the last step is

$$-\frac{3}{2}z = -\frac{3}{2}$$

$0x + 0y + 0z = 1$. This can never hold for any value of x, y, z . Hence, the system has no solution. Remark 2.3.13 Note that to solve a linear system, $Ax = b$, one needs to apply only the elementary row operations to the

augmented matrix $[A \ b]$. 2.4 Row Reduced Echelon Form of a Matrix Definition 2.4.1 (Row Reduced Form of a Matrix) A matrix C is said to be in the row reduced form if 1. the first non-zero entry in each row of C is 1; 2. the column containing this 1 has all its other entries zero. A matrix in the row reduced form is also called a row reduced matrix. Example 2.4.2 1. One of the most important examples of a row

reduced matrix is the $n \times n$ identity matrix, I_n . Recall that the (i, j) th entry of the identity matrix is

δ_{ij} is usually referred to as the Kronecker delta function.

2. The matrices $\square \square \square \begin{smallmatrix} 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{smallmatrix}$

$\square \square \square \begin{smallmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}$ and $\square \square \square \begin{smallmatrix} 0 & 1 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{smallmatrix}$

$\square \square \square \begin{smallmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{smallmatrix}$ are also in row reduced form.

3. The matrix $\square \square \square \begin{smallmatrix} 1 & 0 & 0 & 0 & 5 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{smallmatrix}$

$$\begin{array}{ccc} 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 2 \end{array}$$

$\square \square \square \begin{smallmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}$ is not in the row reduced form. (why?)

Definition 2.4.3 (Leading Term, Leading Column) For a row-reduced matrix, the first non-zero entry of any row is called a leading term. The columns containing the leading terms are called the leading

$$\square \square$$

$$1 \text{ if } i = j$$

$$I_{ij} = \delta_{ij} =$$

$$0 \text{ if } i \neq j.$$

$$\square$$

0 1 1 2

$$\begin{array}{rcl}
 \begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} & \xrightarrow{R_{13}} & \begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} \\
 & \xrightarrow{R_{21}(-2)} & \begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} \\
 & \xrightarrow{R_{23}} & \begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} \\
 & \xrightarrow{R_{32}(2)} & \begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 3 & 3 & 7 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} \\
 & \xrightarrow{R_3(1/3)} & \begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} \\
 & \xrightarrow{R_{12}(-1)} & \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} \\
 & \xrightarrow{R_{23}(-1)} & \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix} \\
 & \xrightarrow{R_{23}(-1)} & \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 2 & 0 & 3 & 5 \\ 0 & -2 & 1 & -1 \end{pmatrix}
 \end{array}$$

Now, consider an $m \times n$ matrix A and an elementary matrix E of order n . Then multiplying by E on the right to A corresponds to applying column transformation on the matrix A . Therefore, for each elementary matrix, there is a corresponding column transformation. We summarize:

Definition 2.4.16 The column transformations obtained by right multiplication of elementary matrices are called elementary column operations.

Example 2.4.17 Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{pmatrix}$

and consider the elementary column operation f which interchanges the second and the third column of A . Then $f(A) = \begin{pmatrix} 1 & 3 & 2 \\ 3 & 2 & 0 \end{pmatrix}$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{pmatrix} = A \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = E_{23}(-1)E_{12}(-1)E_{31}(1/3)E_{32}(2)E_{23}E_{21}(-2)E_{13}(1/3)A.$$

Exercise 2.4.18 1. Let e be an elementary row operation and let $E = e(I)$ be the corresponding elementary matrix. That is, E is the matrix obtained from I by applying the elementary row operations. $0 \ 1 \ 0 = AE_{23}$

Show that $e(A) = EA$.

The (k, H) th entry of $E_i(c)$ is $(E_i)_{(k,0)}$ if $k = H$ and c if $(k, H) = (i, i)$ otherwise. In particular, elementary matrices on the left to the augmented matrix. Does the Gauss-Jordan method also corresponds to multiplying by elementary matrices on the left? Give reasons. 3. Let A and B be two $m \times n$ matrices. Then prove that the two matrices are row-equivalent if and only if $B = PA$, where P is product of elementary matrices. When is this P unique?

2. Show that the Gauss elimination method is same as multiplying by a series of elementary matrices. Bare row-equivalent if and only if $B = PA$, where P is product of elementary matrices.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & c \end{pmatrix} = E_1(c) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & c \end{pmatrix}$$

Example 2.4.15 1. Let $A = \begin{pmatrix} 1 & 2 & 3 & 0 & 2 & 0 & 3 & 4 \\ 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{pmatrix}$

$$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 3 & 4 & 5 & 6 \\ 2 & 0 & 3 & 4 \\ 2 & 0 & 3 & 4 \\ 3 & 4 & 5 & 6 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

A. That is, interchanging the two rows of the matrix A is same as multiplying on the left by the corresponding elementary matrix. In other words, we see that the left multiplication of elementary matrices to a matrix results in elementary row operations.

0 otherwise
c if $i = j = k$

$$E_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2.5 Rank of a Matrix

In previous sections, we solved linear systems using Gauss elimination method or the Gauss-Jordan method. In the examples considered, we have encountered three possibilities, namely 1. existence of a unique solution, 2. existence of an infinite number of solutions, and 3. no solution. Based on the above possibilities, we have the following definition.

Definition 2.5.1 (Consistent, Inconsistent) A linear system is called consistent if it admits a solution. Otherwise, it is called inconsistent.

CHAPTER 2. LINEAR SYSTEM OF EQUATIONS From the last matrix in Step 2b, we deduce $\text{row-rank}(A) = 2$. Remark 2.5.4 Let $Ax = b$ be a linear system with m equations and n unknowns. Then the row-reduced echelon form of A agrees with the first n columns of $[A \ b]$, and hence

is called inconsistent if it admits no solution. The question arises, as to whether there are conditions under which the linear system $Ax = b$ is consistent. The answer to this question is in the affirmative. To proceed further, we need a few definitions and remarks. Recall that the row reduced echelon form of a matrix is unique and therefore, the number of non-zero rows is a unique number. Also, note that the number of non-zero rows in either the row reduced form or the row reduced echelon form of a matrix are same. Definition 2.5.2 (Row rank of a Matrix) The number of non-zero rows in the row reduced form of a matrix is called the row-rank of the matrix. By the very definition, it

is clear that row-equivalent matrices have the same row-rank. For a matrix A , we write 'row-rank (A)' to denote the row-rank of A . The reader is advised to supply a proof. Remark 2.5.5 Consider a matrix A . After application of a finite number of elementary column operations (see Definition 2.4.16) to the matrix A , we can have a matrix, say B , which has the following

rows is a unique number. Also, note that the number of non-zero rows in either the row reduced form or the row reduced echelon form of a matrix are same. Definition 2.5.2 (Row rank of a Matrix) The number of non-zero rows in the row reduced form of a matrix is called the row-rank of the matrix. By the very definition, it

is clear that row-equivalent matrices have the same row-rank. For a matrix A , we write 'row-rank (A)' to denote the row-rank of A .

Example 2.5.3 1. Determine the row-rank of $A = \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 3 & 1 & 2 & 1 & 2 & 1 \end{pmatrix}$

Solution: To determine the row-rank of A , we proceed as follows.

$$\begin{pmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \end{pmatrix} \xrightarrow{R_{21}(-2), R_{31}(-1)} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$$\begin{array}{cccc} R_{23}(-1), R_{13}(1) & \square & \square & \\ 0 & 1 & 1 & \\ 0 & 1 & 0 & \end{array} \quad \begin{array}{ccc} & & \\ & & \square \\ & & 1 \end{array} \begin{array}{ccc} & & \\ & & 0 \\ & & 0 \end{array}$$

2. Determine the row-rank of A

$$\rightarrow R_{21}(-2), R_{31}(-1)$$

Exercise 2.4.12. 2. For any matrix A , prove that $\text{rank}(A) = \text{rank}(A^t)$.

3. Let A be an $n \times n$ matrix with $\text{rank}(A) = n$. Then prove that A is row-equivalent to I_n .

the Gauss-Jordan method reduces to a matrix $[C \ d]$ with

form $I_r B$

2.6.1 Example

used in the proof of the next theorem and is omitted. The interested readers can read the proof in Appendix 14.1.

exactly one of the following statement holds:

1. if $r_1 = r \leq n$, the set of solutions of the linear system is an infinite set and has

$$\{u_0 + k_1 u_1 + k_2 u_2 + \cdots + k_{n-r} u_{n-r} : k_i \in \mathbb{R}, 1 \leq i \leq n-r\}, \text{ where } u_0, u_1, \dots, u_{n-r} \text{ are } n \times 1 \text{ vectors satisfying}$$

3. If $r < r_a$, the linear system has no solution. Remark 2.6.2 Let A be an $m \times n$ matrix and consider the linear system $Ax = b$. Then by

Theorem 2.6.1. we see that the linear system $Ax = b$ is consistent if and only if

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2. The first non-zero entry in the non-zero rows appear in columns 1, 2, 5 and 6.

3. Thus, the respective variables x_1, x_2, x_5 and x_6 are the basic variables.

Let $u_0 =$

, $u_1 =$

0

0

4. The remaining variables, x_3, x_4 and x_7 are free variables.

5. We assign arbitrary constants k_1, k_2 and k_3 to the free variables x_3, x_4 and x_7 , respectively. Hence, we have the set of solutions as

Observations:

$$\begin{aligned} x_1 &= 8 - 2k_1 + k_2 - 2k_3 \\ x_2 &= 1 - k_1 - 3k_2 - 5k_3 \\ x_3 &= k_1 \\ x_4 &= k_2 \\ x_5 &= 4 - k_3 \\ x_6 &= 0 \end{aligned}$$

system $Ax = 0$. Corollary 2.6.3 Let A be an $m \times n$ matrix. Then the homogeneous system $Ax = 0$ has a non-trivial solution if and only if $\text{rank}(A) < n$. Proof. Suppose the system $Ax = 0$ has a non-trivial solution, x_0 . That is, $Ax_0 = 0$ and $x_0 \neq 0$. Under this assumption, we need to show that $\text{rank}(A) < n$. On the contrary, assume that $\text{rank}(A) = n$. So, $n = \text{rank}(A) = \text{rank}[A \ 0] = r_a$.

Also $A0 = 0$ implies that 0 is a solution of the linear system $Ax = 0$. Hence, by the uniqueness of the solution under the condition $r = r_a = n$ (see Theorem 2.6.1), we get $x_0 = 0$. A contradiction to the fact that x_0 was a given non-trivial solution.

Now, let us assume that $\text{rank}(A) < n$. Then

$$r_a = \text{rank}[A \ 0] = \text{rank}(A) < n.$$

So, by Theorem 2.6.1, the solution set of the linear system $Ax = 0$ has infinite number of vectors x satisfying $Ax = 0$. From this infinite set, we can choose any vector x_0 that is different from 0 . Thus, we have a solution $x_0 \neq 0$.

0. That is, we have obtained a non-trivial solution x_0 . We now state another important result whose proof is immediate from Theorem 2.6.1 and Corollary

$$x_7 = k_3$$

$$\begin{pmatrix}
 1 & 0 & 0 & 0 \\
 0 & 1 & 0 & 0 \\
 0 & 0 & 1 & 0 \\
 0 & 0 & 0 & 1
 \end{pmatrix}
 + k_1
 \begin{pmatrix}
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0
 \end{pmatrix}
 + k_2
 \begin{pmatrix}
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0
 \end{pmatrix}
 + k_3
 \begin{pmatrix}
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0
 \end{pmatrix}$$

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Proposition 2.6.4 Consider the linear system $Ax = b$. Then the two statements given below cannot hold together.

1. The system $Ax = b$ has a unique solution for every b .
2. The system $Ax = 0$ has a non-trivial solution.

Remark 2.6.5 1. Suppose x_1, x_2 are two solutions of $Ax = 0$. Then $k_1x_1 + k_2x_2$ is also a solution of $Ax = 0$ for any $k_1, k_2 \in \mathbb{R}$.

36 CHAPTER 2. LINEAR SYSTEM OF EQUATIONS Proof. Note that $C = C I_n = C(AB) = (CA)B = I_n B = B$. Remark 2.7.3 1. From the above lemma, we observe that if a matrix A is invertible, then the inverse is unique. 2. As the inverse of a matrix A is unique, we denote it by A^{-1} . That is, $AA^{-1} = A^{-1}A = I$.

Theorem 2.7.4 Let A and B be two matrices with inverses A^{-1} and B^{-1} , respectively. Then

1. If u, v are two solutions of $Ax = b$ then $u - v$ is a solution of the system $Ax = 0$. That is, $u - v = x_0$ for some solution x_0 of $Ax = 0$. That is, any two solutions of $Ax = b$ differ by a solution of the associated homogeneous system $Ax = 0$.
2. In conclusion, for $b \neq 0$, the set of solutions of the system $Ax = b$ is of the form, $\{x_0 + x_1\}$; where x_0 is a particular solution of $Ax = b$ and x_1 is a solution $Ax = 0$.

3. prove that $(A^t)^{-1} = (A^{-1})^t$.

2.6.3 Exercises Exercise 2.6.6 1. For what values of c and k -the following systems have i) no solution, ii) a unique solution and iii) infinite number of solutions.

(a) $x + y + z = 3, x + 2y + cz = 4, 2x + 3y + 2cz = k$. (b) $x + y + z = 3, x + y + 2cz = 7, x + 2y + 3cz = k$. (c) $x + y + 2z = 3, x + 2y + cz = 5, x + 2y + 4z = k$. (d) $kx + y + z = 1, x + ky + z = 1, x + y + kz = 1$.

(e) $x + 2y - z = 1, 2x + 3y + kz = 3, x + ky + 3z = 2$.

(f) $x - 2y = 1, x - y + kz = 1, ky + 4z = 6$.

2. Find the condition on a, b, c so that the linear system $x + 2y - 3z = a, 2x + 6y - 11z = b, x - 2y + 7z = c$ is consistent.

Proof. Proof of Part 1. By definition $AA^{-1} = A^{-1}A = I$. Hence, if we denote A^{-1} by B , then we get $AB = BA = I$. This again by definition, implies $B^{-1} = A$, or equivalently $(A^{-1})^{-1} = A$. Proof of Part 2. Verify that $(AB)(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})(AB)$. Hence, the result follows by definition. Proof of Part 3. We know $AA^{-1} = A^{-1}A = I$. Taking transpose, we get $(AA^{-1})^t = (A^{-1}A)^t = I^t \Leftrightarrow (A^{-1})^t A^t = A^t (A^{-1})^t = I$. Hence, by definition $(A^{-1})^{-1} = (A^{-1})^t$. Exercise 2.7.5 1. If A is a symmetric matrix, is the matrix A^{-1} symmetric? 2. Show that every elementary matrix is invertible. Is the inverse of an elementary matrix, also an elementary matrix? 3. Let $A_1, A_2, \dots, A_r$ be invertible matrices. Prove that the product $A_1 A_2 \cdots A_r$ is also an invertible

3. Let A be an $n \times n$ matrix. If the system $A^2 x = 0$ has a non trivial solution then show that $Ax = 0$ also has a non trivial solution.

2.7 Invertible Matrices

2.7.1 Inverse of a Matrix Definition 2.7.1 (Inverse of a Matrix) Let A be a square matrix of order n . 1. A square matrix

B is said to be a left inverse of A if $BA = I_n$. 2. A square matrix C is called a right inverse of A , if $AC = I_n$.

1. $(A^{-1})^{-1} = A$.
2. $(AB)^{-1} = B^{-1}A^{-1}$.

matrix. 4. If P and Q are invertible matrices and $P AQ$ is defined then show that $\text{rank}(P AQ) = \text{rank}(A)$. 5. Find

3. A matrix A is said to be invertible (or is said to have an inverse) if there exists a matrix B such

that $AB = BA = I_n$. Lemma 2.7.2 Let A be an $n \times n$ matrix. Suppose that there exist $n \times n$ matrices B and C such that 2.7.

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$B_1 A = R_1 R_2 \begin{pmatrix} 2 & 0 & 0 \end{pmatrix}, AC_1 = S_1 \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}, B_2 AC_2 = A_1 \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$ CHAPTER 2. LINEAR SYSTEM OF EQUATIONS

8. Let A be an $m \times n$ matrix of rank r . Then A can be written as $A = BC$, where both B and C have

matrices P and Q which are product of elementary matrices such that $B = P A Q$ where $A = \begin{pmatrix} 2 & 4 & 8 \end{pmatrix}$

$\begin{pmatrix} 1 & 3 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$

6. Let A and B be two matrices. Show that (a) if $A + B$ is defined, then $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$, (b) if AB is defined, then $\text{rank}(AB) \leq \text{rank}(A)$ and $\text{rank}(AB) \leq \text{rank}(B)$.

7. Let A be any matrix of rank r . Then show that there exists invertible matrices B, C such that $\begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$ and $B_2 AC_2 = I_r \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$. Also, prove that the matrix A_1 is an $r \times r$ invertible matrix.

Suppose A is of full rank. This implies, the row reduced echelon form of A has all non-zero rows. rank r and B is a matrix of size $m \times r$ and C is a matrix of size $r \times n$. 9. Let A and B be two matrices such that AB is

defined and $\text{rank}(A) = \text{rank}(AB)$. Then show that

But A has as many columns as rows and therefore, the last row of the row reduced echelon form of A will be $(0, 0, \dots, 0, 1)$.

Hence, the row reduced echelon form of A is the identity matrix. $3 \Rightarrow 4$

$$\begin{pmatrix} 0 & 0 & 4 & -1 \\ 3 & -1 & 3 & -3 \\ 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{pmatrix} \xrightarrow{R_3(3/4)} \begin{pmatrix} 0 & 0 & 4 & -1 \\ 3 & -1 & 3 & -3 \\ 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

$AB = A[x_1, x_2, \dots, x_n] = [Ax_1, Ax_2, \dots, Ax_n] = [e_1, e_2, \dots, e_n] = I_n$. Therefore, by Theorem 2.7.8, the matrix A is invertible.

Exercise 2.7.11 1. Show that a triangular matrix A is invertible if and only if each diagonal entry of A

$$\begin{pmatrix} 0 & 1 & 1 & 3 \\ 3 & 0 & 0 & 1 \\ 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & 3 \end{pmatrix} \xrightarrow{R_3(3/4)} \begin{pmatrix} 0 & 1 & 1 & 3 \\ 3 & 0 & 0 & 1 \\ 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

is non-zero.

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 1 & 0 & -1 \end{pmatrix} \xrightarrow{R_3(3/4)} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

0 1
invertible? Give reasons.

3. Let A be an $n \times m$ matrix and B be an $m \times n$ matrix. Prove that the matrix $I - BA$ is invertible if and only if the matrix $I - AB$ is

We first give a consequence of Theorem

invertible. 2.7.3 Inverse and Gauss-Jordan Method

2.7.7 and then use it to find the inverse of an invertible matrix-

Corollary 2.7.12 Let A be

an invertible $n \times n$ matrix. Suppose that a sequence of elementary row-operations reduces A to the identity matrix. Then the same sequence of elementary row-operations when applied to the identity matrix yields A^{-1} .

$$\begin{pmatrix} 0 & 0 & 1 & -1 \\ 1 & -1 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{pmatrix} \xrightarrow{R_1(4)} \begin{pmatrix} 0 & 0 & 1 & -1 \\ 1 & -1 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{pmatrix}$$

Proof. Let A be a square matrix of order n. Also, let $E_1, E_2, \dots, E_k$ be a sequence of elementary row operations such that $E_k E_{k-1} \dots E_1 A = I_n$. Then $E_k E_{k-1} \dots E_1 I_n = A^{-1}$. This implies $A^{-1} = E_k E_{k-1} \dots E_1 A$.

Notation: For an $n \times n$ matrix A, by $A(\alpha|\beta)$, we mean the submatrix Bof A, which is obtained by deleting the α^{th} row and β^{th} column.

Example 2.8.1 Consider a matrix $A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$

Summary: Let A be an $n \times n$ matrix. Apply the Gauss-Jordan method to the matrix $[A \ I_n]$. Suppose the row reduced echelon form A is not invertible.

Example 2.7.13 Find the inverse of the matrix $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}$

1 2 using Gauss-Jordan method. 1 2 1

A(1, 2|1, 3) = [4]. Definition 2.8.2 (Determinant of a Square Matrix)

Let A be a square matrix of order n. With A, we associate inductively (on n) a number, called the determinant of A, written $\det(A)$ (or $|A|$)

by $\det(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

Solution: Consider the matrix $\begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \end{pmatrix}$

1 1 2 0 0 1. A sequence of steps in the Gauss-Jordan method

2.8. DETERMINANT 41

Definition 2.8.3 (Minor, Cofactor of a Matrix) The number $\det(A(i|j))$ is called the $(i, j)^{th}$ minor of A. We write A_{ij}

= $\det(A(i|j))$. The $(i, j)^{th}$ cofactor of A, denoted C_{ij} , is the number $(-1)^{i+j} A_{ij}$. Example 2.8.4 1. Let $A = \begin{pmatrix} a_{11} & a_{12} \end{pmatrix}$

$$\det(A) = \begin{vmatrix} a \end{vmatrix} = a \text{ if } A = [a] \text{ (} n = 1 \text{),}$$

(i)

$$\det(A) = a^n$$

42 CHAPTER 2. LINEAR SYSTEM OF EQUATIONS Remark 2.8.8 1. Many authors define the determinant using "Permutations." It

2. Let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

are: $\det(A) = |A| = a_{11}a_{22}a_{33} - a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$

$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22}) = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} \quad (2.8.1)$$

For example, if $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix}$

1 2 then

$$\det(A) = |A| = 1 \cdot \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 1 \cdot (3 \cdot 2 - 1 \cdot 1) = 5$$

$$2 \cdot \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = 2 \cdot (2 \cdot 2 - 1 \cdot 3) = 2 \cdot 1 = 2$$

$$3 \cdot \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} = 3 \cdot (2 \cdot 1 - 3 \cdot 3) = 3 \cdot (-7) = -21$$

$$= 5 + 2 - 21 = -14$$

Exercise 2.8.5 1. Find the determinant of the following matrices.

$$\begin{pmatrix} 0 & 0 & 5 \\ 1 & 2 & 7 \\ 8 & 2 & 1 \end{pmatrix}, \text{ (ii) } \begin{pmatrix} 3 & 5 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} a_{21} & a_{22} \end{pmatrix}, \text{ (iii) } \begin{pmatrix} 1 & a & a^2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 3 \\ 0 & 4 & 3 \\ 0 & 2 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & b & b^2 \\ 0 & 2 & 3 \\ 6 & -7 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

$$\begin{pmatrix} 1 & c \end{pmatrix}$$

2.

Let A be a 1×2 matrix and B be a 2×1 matrix having positive entries. Which of BA or AB is

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 4 & -1 \\ -1 & 4 \end{pmatrix}$$

2.

$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 3 & 4 \\ 4 & -1 & 4 \end{pmatrix}$$

2.

$$\begin{pmatrix} 3/4 & -1/4 & -1/4 \\ -1/4 & 3/4 & -1/4 \end{pmatrix}$$

Exercise 2.7.14 Find the inverse of the following matrices using Gauss-Jordan method.

$$R_{31}(-1)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \end{pmatrix}, \text{ (ii) } \begin{pmatrix} 1 & 3 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 & 7 \end{pmatrix}, \text{ (iii) } \begin{pmatrix} 2 & -1 & 3 \\ -1 & 3 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 & 1 \end{pmatrix}$$

2.8 Determinant

4.

of the matrix $[A \ I_n]$ is $[B \ C]$. If $B = I_n$, then $A^{-1} = C$ or else

$$\begin{pmatrix} 2 & 4 & 7 \end{pmatrix}. \text{ Then } A(1|2) = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}, A(1|3) = \begin{pmatrix} 1 & 3 \end{pmatrix}, \text{ and } \begin{pmatrix} 1 & 3 \end{pmatrix}$$

4.

$$R_{23} \xrightarrow{(-1/3)} R_{13}(-1/2) \begin{pmatrix} 1 & 1/2 & 5/8 & -3/8 \end{pmatrix}$$

turns out that the way we have defined determinant is usually called the expansion of the determinant along the first row. 2. Part 1 of

Lemma 2.8.7 implies that "one can also calculate the determinant by expanding along any row." Hence, for an $n \times n$ matrix A, for every k , $1 \leq k \leq n$, one also has

$$\det(A) = |A| = a_{11}A_{11} - a_{12}A_{12} = a_{11}a_{22} - a_{12}a_{21}. \text{ For example, for } A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \det(A) = |A| = 1 \cdot 2 - 2 \cdot 2 = -3.$$

2.8. DETERMINANT 45 Step 2. Suppose $\det(A) = 0$. Then A is not invertible. Hence, there exists an invertible matrix P such that $PA =$

C, where $C = C \begin{pmatrix} 1 & \\ & 0 \end{pmatrix}$. So, $A = P^{-1}C$, and therefore

$$\begin{aligned} \det(AB) &= \det((P^{-1}C)B) = \det(P^{-1}(CB)) = \det P^{-1} \det C \det B \\ &= \det(P^{-1}) \cdot \det C \det B \\ &= \det(P)^{-1} \cdot \det C \det B \end{aligned}$$

Since P^{-1} is non-singular, $\det(P)^{-1} \neq 0$. Thus, the proof of the theorem is complete.

Corollary 2.8.16 Let A be a square matrix. Then A is non-singular if and only if A has an inverse.

$$\det(A) = \det(A^{-1})^{-1}$$

The next corollary is an easy consequence of Theorem 2.8.12 (recall Theorem 2.7.8).

by using Parts 1, 2 and 4 of Lemma 2.8.7 repeatedly, we get $\det(AB) = \det(E_1 E_2 \cdots E_k B) = \det(E_1) \det(E_2) \cdots \det(E_k) \det(B)$

$$\begin{aligned} &= \det(E_1) \det(E_2) \det(E_3) \cdots \det(E_k) \det(B) \\ &= \det(E_1 E_2) \det(E_3) \cdots \det(E_k) \det(B) \\ &= \det(E_1 E_2 \cdots E_k) \det(B) \end{aligned}$$

Thus, we get the required result in case A is non-singular.

$a_{ij} C_{ik} = \det(A)$ if $j = k$

46 CHAPTER 2. LINEAR SYSTEM OF EQUATIONS Proof. Since $\det(A) \neq 0$, $A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$. Thus, the linear system $Ax = b$ has the solution $x = \frac{1}{\det(A)} \text{Adj}(A)b$. Hence, x_j , the j th coordinate of x is given by $x_j = b_1 C_{1j} + b_2 C_{2j} + \cdots + b_n C_{nj}$

$$\det(A) x_j = b_1 a_{1j} + b_2 a_{2j} + \cdots + b_n a_{nj}$$

The theorem implies that

Corollary 2.8.14 If A is a non-singular matrix, then

$\text{Adj}(A) A = \det(A) I_n$ and $A \text{Adj}(A) = \det(A) I_n$

Proof. Step 1. Let A and B be square matrices of order n . Then $\det(AB) = \det(A) \det(B)$.

Suppose A has an inverse. Then there exists a matrix B such that $AB = I = BA$. Taking determinant of both sides, we get

$$\det(A) \det(B) = \det(I) = 1$$

This means, A is invertible. Therefore, either A is an elementary matrix or is a product of elementary matrices (see Theorem 2.7.7). So, let $E_1, E_2, \dots, E_k$ be elementary matrices such that $A = E_1 E_2 \cdots E_k$. Then,

$b_1 a_{12} \cdots a_{1n}$
 $b_2 a_{22} \cdots a_{2n}$
 $\vdots$
 $b_n a_{n2} \cdots a_{nn}$, and in general

Proof. Suppose A is non-singular. Then $\det(A) \neq 0$ and therefore, $A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$. Thus, A has an inverse.

Suppose A has an inverse. Then there exists a matrix B such that $AB = I = BA$. Taking determinant of both sides, we get

$$a_{11} \cdots a_{1j-1} b_1 a_{1j+1} \cdots a_{1n} a_{21} \cdots a_{2j-1} b_2 a_{2j+1} \cdots a_{2n} a_{31} \cdots a_{3j-1} b_3 a_{3j+1} \cdots a_{3n} \cdots a_{n1} \cdots a_{nj-1} b_n a_{nj+1} \cdots a_{nn} = \det(A)$$

$\det(A) \det(B) = \det(AB) = \det(I) = 1$. This implies that $\det(A) \neq 0$. Thus, A is non-singular.

Theorem 2.8.17 Let A be a square matrix. Then $\det(A) = \det(A^t)$.

Proof. If A is a non-singular Corollary 2.8.14 gives $\det(A) = \det(A^t)$.

If A is singular, then $\det(A) = 0$. Hence, by Corollary 2.8.16, A doesn't have an inverse. Therefore, A^t also doesn't have an inverse (for if A^t has an inverse then $A^{-1} = (A^t)^{-t}$). Thus again by Corollary 2.8.16, $\det(A^t) = 0$.

Therefore, we again have $\det(A) = 0 = \det(A^t)$. Hence, we have $\det(A) = \det(A^t)$.

0

0

2.8.2 Cramer's Rule

Recall the following:

- The linear system $Ax = b$ has a unique solution for every b if and only if A^{-1} exists.
- A has an inverse if and only if $\det(A) \neq 0$.

for $j = 2, 3, \dots, n$.

Example 2.8.19 Suppose that $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 1 \\ 2 & 3 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Use Cramer's rule to find a vector x such that $Ax = b$.

Solution: Check that $\det(A) = 1$. Therefore $x_1 = 1$.

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 2 & 1 \\ 2 & 3 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} - 2 \cdot \begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} + 3 \cdot \begin{vmatrix} 2 & 2 \\ 2 & 2 \end{vmatrix} = 1 \cdot (-1) - 2 \cdot 0 + 3 \cdot 0 = -1$$

$x_2 = 1$, and $x_3 = 1$.

$x^t = (-1, 1, 0)$.

$$x_1 = \frac{\det(A_1)}{\det(A)}$$

2.9 Miscellaneous Exercises

Exercise 2.9.1 1. Let A be an orthogonal matrix. Show that $\det A = \pm 1$.

2. If A and B are two $n \times n$ non-singular matrices, are the matrices $A+B$ and $A-B$ non-singular? Justify your answer.

3. For an $n \times n$ matrix A , prove that the following conditions are equivalent: $x_j = \frac{1}{\det(A)}$

Thus, $Ax = b$ has a unique solution for every b if and only if $\det(A) \neq 0$. The following theorem gives a direct method of finding the solution of the linear system $Ax = b$ when $\det(A) \neq 0$.

Theorem 2.8.18 (Cramer's Rule) Let $Ax = b$ be a linear system with n equations in n unknowns.

$\det(A) \neq 0$, then the unique solution to this system is

$\det(A_j)$ for $j = 1, 2, \dots, n$, where A_j is the matrix obtained from A by replacing the j th column of A by the column vector b . $x_j = \frac{\det(A_j)}{\det(A)}$

(a) A is singular (A^{-1} doesn't exist).
(b) $\text{rank}(A) < n$.
(c) $\det(A) = 0$.
(d) A is not row-equivalent to I_n , the identity matrix of order n .
(e) $Ax = 0$ has a non-trivial solution for x .

that $Ax = b$.
(f) $Ax = b$ doesn't have a unique solution, i.e., it has no solutions or it has infinitely many solutions.

2.9 MISCELLANEOUS EXERCISES

47 2 0 6 0 4
5 3 2 2 7
4. Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$. We know that the numbers 20604, 53227, 25755, 20927 and 78421 are

2 0 9 2 7
all divisible by 17. Does this imply 17 divides $\det(A)$?
5. Let $A = [a_{ij}]_{n \times n}$ where $a_{ij} = x^{i+j}$

1. Show that $\det(A) = 0$.
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($x_1 - x_i$). [The matrix A is usually called the Van-dermonde matrix.]
6. Let $A = [a_{ij}]$ with $a_{ij} = \max\{i, j\}$ be an $n \times n$ matrix. Compute $\det A$.
7. Let $A = [a_{ij}]$ with $a_{ij} = 1/(i+j)$ be an $n \times n$ matrix. Show that A is invertible.

8. Solve the following system of equations by Cramer's rule.
i) $x + y + z - w = 1$, $x + y - z + w = 2$, $2x + y + z - w = 7$, $x + y + z + w = 3$.
ii) $x - y + z - w = 1$, $x + y - z + w = 2$, $2x + y - z - w = 7$, $x - y - z + w = 3$.
9. Suppose $A = [a_{ij}]$ and $B = [b_{ij}]$ are two $n \times n$ matrices such that $b_{ij} = p^{-1} a_{ij}$ for $1 \leq i, j \leq n$ for some non-zero real number p . Then compute $\det(B)$ in terms of $\det(A)$.
10. The position of an element a_{ij} of a determinant is called even or odd according as $i+j$ is even or odd. Show that

3. **Distributive Laws:** relating vector addition with scalar multiplication. For any $\alpha, \beta \in F$ and $u, v \in V$, the following distributive laws hold:

52 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES7. Consider the set $C^n = \{(z_1, z_2, \dots, z_n) : z_i \in \mathbb{C} \text{ for } 1 \leq i \leq n\}$. For $(z_1, \dots, z_n), (w_1, \dots, w_n) \in C^n$, $(z_1, \dots, z_n) \oplus (w_1, \dots, w_n) = (z_1 + w_1, \dots, z_n + w_n)$ and $\alpha(z_1, \dots, z_n) = (\alpha z_1, \dots, \alpha z_n)$.

(a) If the set F is the set C of complex numbers, then C^n is a complex vector space having n -tuple of complex numbers as its vectors. (b) If the set F is the set R of real numbers, then C^n is a real vector space having n -tuple of complex numbers as its vectors. Remark 3.1.5 In Example 7a, the scalars are Complex numbers and hence $i(1, 0) = (i, 0)$. Whereas, in Example 7b, the scalars are Real Numbers and hence we cannot write $i(1, 0) =$

3.1.2 Examples Example 3.1.4 1. The set R of real numbers, with the usual addition and multiplication (i.e., $+$ and $\cdot$) forms a vector space over R .

2. Consider the set $R^2 = \{(x_1, x_2) : x_1, x_2 \in R\}$. For $x_1, x_2, y_1, y_2 \in R$ and $\alpha \in R$, define, $(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$ and $\alpha \$ (x_1, x_2) = (\alpha x_1, \alpha x_2)$.

Then R^2 is a real vector space.

3. Let $R^n = \{(a_1, a_2, \dots, a_n) : a_i \in R, 1 \leq i \leq n\}$, be the set of n -tuples of real numbers. For $u = (a_1, \dots, a_n)$, $v = (b_1, \dots, b_n)$ in V and $\alpha \in R$, we define

$$u \oplus v = (a_1 + b_1, \dots, a_n + b_n) \text{ and } \alpha \$ u = (\alpha a_1, \dots, \alpha a_n)$$

(called component wise or coordinate wise operations). Then V is a real vector space with addition and scalar multiplication defined as above. This vector space is denoted by R^n , called the real vector space of n -tuples.

4. Let $V = R^+$ (the set of positive real numbers). This is not a vector space under usual operations of addition and scalar multiplication (why?). We now define a new vector addition and scalar multiplication for all v_1, v_2, v

$\in R^+$ and $\alpha \in R$. Then R^+ is a real vector space with 1 as the additive identity.

8. Fix a positive integer n and let $M_n(R)$ denote the set of all $n \times n$ matrices with real entries. Then $M_n(R)$ is a real vector space with vector addition and scalar multiplication defined by $A \oplus B = [a_{ij}] \oplus [b_{ij}] = [a_{ij} + b_{ij}]$, $\alpha \$ A$

$= \alpha [a_{ij}] = [\alpha a_{ij}]$. 9. Fix a positive integer n . Consider the set, $P_n(R)$, of all polynomials of degree $\leq n$ with coefficients from R in the indeterminate x . Algebraically,

$$P_n(R) = \{a_0 + a_1x + a_2x^2 + \dots + a_nx^n : a_i \in R, 0 \leq i \leq n\}.$$

and $\alpha \in F$, define,

Let $f(x), g(x) \in P_n(R)$. Then $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ and $g(x) = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$ for some $a_i, b_i \in R, 0 \leq i \leq n$. It can be verified that $P_n(R)$ is a real vector space with the addition and scalar multiplication

defined by: $f(x) \oplus g(x) = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n$, and

5. Let $V = R^2$. Define $(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1 + 1, x_2 + y_2 - 3)$, $\alpha \$ (x_1, x_2) = (\alpha x_1 + \alpha -$

$\alpha \$ f(x) = \alpha a_0 + \alpha a_1x + \dots + \alpha a_nx^n$ for $\alpha \in R$.

1. $\alpha x_2 - 3\alpha + 3$) for $(x_1, x_2), (y_1, y_2) \in R^2$ and $\alpha \in R$. Then it can be easily verified that the vector $(-1, 3)$ is the additive

identity and V is indeed a real vector space.

Recall $\sqrt{-1}$ is denoted i .

6. Consider the set $C = \{x + iy : x, y \in R\}$ of complex numbers. (a) For $x_1 + iy_1, x_2 + iy_2 \in C$ and $\alpha \in R$, define,

$$(x_1 + iy_1) \oplus (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2) \text{ and } \alpha \$ (x_1 + iy_1) = (\alpha x_1) + i(\alpha y_1).$$

Then C is a real vector space.

(b) For $x_1 + iy_1, x_2 + iy_2 \in C$ and $\alpha + i\beta \in C$, define,

$$(x_1 + iy_1) \oplus (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2) \text{ and } (\alpha + i\beta) \$ (x_1 + iy_1) = (\alpha x_1 - \beta y_1) + i(\alpha y_1 + \beta x_1).$$

as

Then C forms a complex vector space.

$$v_1 \oplus v_2 = v_1 + v_2 \text{ and } \alpha \$ v = v^\alpha$$

real vector space. The readers are advised to justify the statements made in the above examples. From now on, we will use 'u +

v' in place of 'u $\oplus$ v' and ' $\alpha \cdot u$ or αu ' in place of ' $\alpha \$ u$ '.

10. Consider the set $P(R)$, of all polynomials with real coefficients. Let $f(x), g(x) \in P(R)$. Observe that a polynomial of the

form $a_0 + a_1x + \dots + a_mx^m$ can be written as $a_0 + a_1x + \dots + a_mx^m + 0 \cdot x^{m+1} + \dots + 0 \cdot x^p$ for any $p > m$. Hence, we can

assume $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_px^p$ and $g(x) = b_0 + b_1x + b_2x^2 + \dots + b_px^p$ for some $a_i, b_i \in R, 0 \leq i \leq p$, for some

large positive integer p . We now define the vector addition and scalar multiplication as

(i, 0).

$$f(x) \oplus g(x) = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_p + b_p)x^p, \text{ and } \alpha \$ f(x) = \alpha a_0 + \alpha a_1x + \dots + \alpha a_px^p \text{ for } \alpha \in R.$$

Then $P(R)$ forms a real vector space.

11. Let $C([-1, 1])$ be the set of all real valued continuous functions on the interval $[-1, 1]$.

For $f, g \in C([-1, 1])$, $(f \oplus g)(x) = f(x) + g(x)$, and

$(\alpha \$ f)(x) = \alpha f(x)$, for all $x \in [-1, 1]$.

Then $C([-1, 1])$ forms a real vector space. The operations defined above are called point wise addition and scalar

multiplication. 54 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES(e) $\{(x_1, x_2, \dots, x_n) : \text{either } x_1 \text{ or } x_2 \text{ or both is } 0\}$.

(f) $\{(x_1, x_2, \dots, x_n) : |x_i| \leq 1\}$.

3. Which of the following are subspaces of $i)C^0(R)$ ii) $C^1(R)$?

(a) $\{(z_1, z_2, \dots, z_n) : z_i \text{ is real}\}$.

(b) $\{(z_1, z_2, \dots, z_n) : z_1 + z_2 = z_3\}$.

$C([-1, 1])$ and $\alpha \in R$, define

(c) $\{(z_1, z_2, \dots, z_n) : |z_1| = |z_2|\}$.

3.1. VECTOR SPACES 53 12. Let V and W be real vector spaces with binary operations $(+, \cdot)$ and $(\oplus, \$)$, respectively. Consider the

following operations on the set $V \times W$: for $(x_1, y_1), (x_2, y_2) \in V \times W$ and $\alpha \in R$, define $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 \oplus y_2)$, and

$\alpha \cdot (x_1, y_1) = (\alpha \cdot x_1, \alpha \$ y_1)$. On the right hand side, we write $x_1 + x_2$ to mean the addition in V , while $y_1 \oplus y_2$ is the addition in W .

Similarly, $\alpha \cdot x_1$ and $\alpha \$ y_1$ come from scalar multiplication in V and W , respectively. With the above definitions, $V \times W$ also forms a

3.1.3 Subspaces

Definition 3.1.6 (Vector Subspace) Let S be a non-empty subset of V . $S(F)$ is said to be a subspace of $V(F)$ if $\alpha u + \beta v \in S$ whenever $\alpha, \beta \in F$ and $u, v \in S$; where the vector addition and scalar multiplication are the

same as that of $V(F)$. Remark 3.1.7 Any subspace is a vector space in its own right with respect to the vector addition and scalar multiplication that is defined for $V(F)$.

Example 3.1.8 1. Let $V(F)$ be a vector space.

Then

(a) $S = \{0\}$, the set consisting of the zero vector 0,

are vector subspaces of V . These are called trivial subspaces.

2. Let $S = \{(x, y, z) \in R^3 : x + y - z = 0\}$. Then S is a subspace of R^3 . (S is a plane in R^3 passing through the origin.)

3. Let $S = \{(x, y, z) \in R^3 : x + y + z = 3\}$. Then S is not a subspace of R^3 . (S is again a plane in R^3 but it doesn't pass through the origin.)

4. Let $S = \{(x, y, z) \in R^3 : z = x\}$. Then S is a subspace of R^3 .

5. The vector space $P_n(R)$ is a subspace of the vector space $P(R)$.

Exercise 3.1.9 1. Which of the following are correct statements?

(a) Let $S = \{(x, y, z) \in R^3 : z = x^2\}$. Then S is a subspace of R^3 .

(b) Let $V(F)$ be a vector space. Let $x \in V$. Then the set $\{\alpha x : \alpha \in F\}$ forms a vector subspace of V . (c) Let $W = \{f \in C([-1, 1]) : f(1/2) = 0\}$. Then W is a subspace of the real vector space.

2. Which of the following are subspaces of $R^3(R)$?

(a) $\{(x_1, x_2, \dots, x_n) : x_i \geq 0\}$.

(b) $\{(x_1, x_2, \dots, x_n) : x_1 + 2x_2 = 4x_3\}$.

(c) $S = V$

(d) $\{(x_1, x_2, \dots, x_n) : x_i \text{ is rational}\}$.

(e) $\{(x_1, x_2, \dots, x_n) : x_1 = x^2\}$.

$C([-1, 1])$.

3.1.4 Linear Combinations Definition 3.1.10 (Linear Span) Let $V(F)$ be a vector space and let $S = \{u_1, u_2, \dots, u_n\}$ be a non-empty subset of V . The linear span of S is the set defined by $L(S) = \{\alpha_1 u_1 + \alpha_2 u_2 + \dots +$

$$\alpha_n u_n : \alpha_i \in F, 1 \leq i \leq n\}$$

If S is an empty set we define $L(S) = \{0\}$.

Example 3.1.11 1. Note that $(4, 5, 5)$ is a linear combination of $(1, 0, 0)$, $(1, 1, 0)$, and $(1, 1, 1)$ as $(4, 5, 5) = 5(1, 1, 1) - 1(1, 0, 0) + 0(1, 1, 0)$.

For each vector, the linear combination in terms of the vectors $(1, 0, 0)$, $(1, 1, 0)$, and $(1, 1, 1)$ is unique.

2. Is $(4, 5, 5)$ a linear combination of $(1, 2, 3)$, $(-1, 1, 4)$ and $(3, 3, 2)$? Solution: We want to find $\alpha_1, \alpha_2, \alpha_3 \in R$ such that

$$\alpha_1(1, 2, 3) + \alpha_2(-1, 1, 4) + \alpha_3(3, 3, 2) = (4, 5, 5). \quad (3.1.1)$$

Check that $3(1, 2, 3) + (-1)(-1, 1, 4) + 0(3, 3, 2) = (4, 5, 5)$. Also, in this case, the vector $(4, 5, 5)$ does not have a unique expression as linear combination of vectors

$(1, 2, 3)$, $(-1, 1, 4)$ and $(3, 3, 2)$. Verify that $(4, 5, 5)$ is not a linear combination of the vectors $(1, 2, 1)$ and $(1, 1, 0)$? 4. The linear span of $S = \{(1, 1, 1), (2, 1, 3)\}$ over R is $L(S) = \{\alpha(1, 1, 1) + \beta(2, 1, 3) : \alpha, \beta \in R\}$

$$= \{(\alpha + 2\beta, \alpha + \beta, \alpha + 3\beta) : \alpha, \beta \in R\}$$

$$= \{(x, y, z) \in R^3 : 2x - y = z\}.$$

as $2(\alpha + 2\beta) - (\alpha + \beta) = \alpha + 3\beta$, and if $z = 2x - y$, take $\alpha = 2y - x$ and $\beta = x - y$.

Lemma 3.1.12 (Linear Span is a subspace) Let $V(F)$ be a vector space and let S be a non-empty subset of V . Then $L(S)$ is a subspace of $V(F)$.

Proof. By definition, $S \subset L(S)$ and hence $L(S)$ is non-empty subset of V . Let $u, v \in L(S)$. Then, for $1 \leq i \leq n$ there exist vectors $w_i \in S$, and scalars $\alpha_i, \beta_i \in F$ such that $u = \alpha_1 w_1 + \alpha_2 w_2 + \dots + \alpha_n w_n$ and $v = \beta_1 w_1 + \beta_2 w_2 + \dots +$

$\beta_n w_n$. Hence,

$$u + v = (\alpha_1 + \beta_1)w_1 + \dots + (\alpha_n + \beta_n)w_n \in L(S).$$

Thus, $L(S)$ is a vector subspace of $V(F)$.

$(3, 3, 2)$.

3.1. VECTOR SPACES 55 Remark 3.1.13 Let V (F) be a vector space and $W \subset V$ be a subspace. If $S \subset W$, then $L(S) \subset W$ is a

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subspace of W as W is a vector space in its own right.

Theorem 3.1.14 Let S be a non-empty subset of a vector space V . Then $L(S)$ is the smallest subspace of V containing S . Proof. For every $u \in S$, $u = 1 \cdot u \in L(S)$ and therefore, $S \subseteq L(S)$. To show $L(S)$ is the smallest

subspace of V containing S , consider any subspace W of V containing S . Then by Proposition 3.1.13,

Proof. Part 1) can be easily proved. Let A be an $m \times n$ matrix. For part 2), let D be the row-reduced form of A with non-zero rows $d_1^t, d_2^t, \dots, d_r^t$. Then $B = E_k E_{k-1} \cdots E_2 E_1 A$ for some elementary matrices $E_1, E_2, \dots, E_k$.

Then, a repeated application of Lemma 3.1.16 implies $\text{Row Space}(A) = \text{Row Space}(B)$. That is, if the rows of the matrix A are $a_1^t, a_2^t, \dots, a_m^t$, then $L(a_1, a_2, \dots, a_m) = L(b_1, b_2, \dots, b_r)$. Hence the required result follows.

Exercise 3.1.18 1. Show that any two row-equivalent matrices have the same row space. Give examples to show that the column space of two row-equivalent matrices need not be the same. 2. Find all the vector subspaces

of $\mathbb{R}^2$.

$L(S) \subseteq W$ and hence the result follows. Definition 3.1.15 Let A be an $m \times n$ matrix with real entries. Then using the rows $a_1^t, a_2^t, \dots, a_m^t \in \mathbb{R}^n$

and columns $b_1, b_2, \dots, b_n \in \mathbb{R}^m$, we define

1. $\text{RowSpace}(A) = L(a_1, a_2, \dots, a_m)$,

2. $\text{ColumnSpace}(A) = L(b_1, b_2, \dots, b_n)$,

4. $\text{Range}(A)$, denoted $\text{Im}(A) = \{y : Ax = y \text{ for some } x^t \in \mathbb{R}^n\}$. Note that the "column space" of a matrix A consists of all b such that $Ax = b$ has a solution. Hence, $\text{ColumnSpace}(A) = \text{Range}(A)$. Lemma 3.1.16 Let A be a

real $m \times n$ matrix. Suppose $B = EA$ for some elementary matrix E . Then $\text{Row Space}(A) = \text{Row Space}(B)$. V containing S .

3. Let P and Q be two subspaces of a vector space V . Show that $P \cap Q$ is a subspace of V . Also show that $P \cup Q$ need not be a subspace of V . When is $P \cup Q$ a subspace of V ? 4. Let P and Q be two subspaces of a vector space V . Define $P + Q = \{u + v : u \in P, v \in Q\}$. Show that $P + Q$ is a subspace of V . Also show that $L(P \cup Q) = P + Q$. 5. Let $S = \{x_1, x_2, x_3, x_4\}$ where $x_1 = (1, 0, 0, 0)$, $x_2 = (1, 1, 0, 0)$, $x_3 = (1, 2, 0, 0)$, $x_4 = (1, 1, 1, 0)$.

Determine all x such that $L(S) = L(S \setminus \{x\})$. 6. Let $C([-1, 1])$ be the set of all continuous functions on the interval $[-1, 1]$ (cf. Example 3.1.4.11). Let

$W_1 = \{f \in C([-1, 1]) : f(0.2) = 0\}$, and

$W_2 = \{f \in C([-1, 1]) : f(\frac{1}{4}) \text{ exists}\}$.

Are W_1, W_2 subspaces of $C([-1, 1])$?

Proof. We prove the result for the elementary matrix $E_i(c)$, where $c \neq 0$ and $i < j$. Let $a_1^t, a_2^t, \dots, a_m^t$

be the rows of the matrix A . Then $B = E_i(c)A$ gives us $\text{Row Space}(B) = L(a_1, \dots, a_{i-1}, a_i + ca_j, \dots, a_m) = \{a_1 a_1 + \dots + a_{i-1} a_{i-1} + a_i(a_i$

$+ ca_j) + \dots$

$= L(a_1, \dots, a_{i-1}, a_i, \dots, a_m)$

$= \text{Row Space}(A)$

Theorem 3.1.17 Let A be an $m \times n$ matrix with real entries. Then

3. $\text{NullSpace}(A)$, denoted $N(A)$ as $\{x^t \in \mathbb{R}^n : Ax = 0\}$. 1. $N(A)$ is a subspace of $\mathbb{R}^n$;

2. the non-zero row vectors of a matrix in row-reduced form, forms a basis for the row-space. Hence $\dim(\text{Row Space}(A)) = \text{rank of } (A)$.

$+ a_m a_m : a_0 \in \mathbb{R}, 1 \leq i \leq m\}$

$= \{a_0 a_0 + a_i ca_j : a_0 \in \mathbb{R}, 1 \leq i \leq m$

$= a_m$

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$\beta_0 a_0 : \beta_0 \in \mathbb{R}, 1 \leq i \leq m$

$= a_m$

3.2 Linear Independence

Definition 3.2.1 (Linear Independence and Dependence) Let $S = \{u_1, u_2, \dots, u_m\}$ be any non-empty subset of V . If there exist some

non-zero α_i 's $1 \leq i \leq m$, such that $\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_m u_m = 0$,

7. Let $V = \{(x, y) : x, y \in \mathbb{R}\}$ over $\mathbb{R}$. Define $(x, y) \oplus (x_1, y_1) = (x + x_1, 0)$ and $\alpha(x, y) = (\alpha x, 0)$. Show that V is not a vector

then the set S is called a linearly dependent set. Otherwise, the set S is called linearly independent.

Example 3.2.2 1. Let $S = \{(1, 2, 1), (2, 1, 4), (3, 3, 5)\}$. Then check that $1(1, 2, 1) + 1(2, 1, 4) + (-1)(3, 3, 5) =$

$(0, 0, 0)$. Since $\alpha_1 = 1, \alpha_2 = 1$ and $\alpha_3 = -1$ is a solution of (3.2.1), so the set S is a linearly dependent subset of $\mathbb{R}^3$.

$L(u_1, u_2, \dots, u_k) = L(u_1, u_2, \dots, u_{k-1})$. The next corollary follows immediately from Theorem 3.2.4 and Corollary 3.2.5.

2. Let $S = \{(1, 1, 1), (1, 1, 0), (1, 0, 1)\}$. Suppose there exists $\alpha, \beta, \gamma \in \mathbb{R}$ such that $\alpha(1, 1, 1) + \beta(1, 1, 0) + \gamma(1, 0, 1) = (0, 0, 0)$. Then check that in this case we necessarily have $\alpha = \beta = \gamma = 0$ which shows that the set $S = \{(1, 1, 1), (1, 1, 0), (1, 0, 1)\}$ is a linearly independent subset of $\mathbb{R}^3$. In other words, if $S = \{u_1, u_2, \dots, u_m\}$ is a non-empty subset of a vector space V , then to

check whether the set S is linearly dependent or independent, one needs to consider the equation $\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_m u_m = 0$. (3.2.1)

In case $\alpha_1 = \alpha_2 = \dots = \alpha_m = 0$ is the only solution of (3.2.1), the set S becomes a linearly independent subset of V . Otherwise, the set S

becomes a linearly dependent subset of V . Proposition 3.2.3 Let V be a vector space. 1. Then the zero-vector cannot belong to a

linearly independent set. 2. If S is a linearly independent subset of V , then every subset of S is also linearly independent. 3. If S is a

linearly dependent subset of V then every set containing S is also linearly dependent. Proof. We give the proof of the first part. The reader is required to supply the proof of other parts,

exists a vector $v \in V$, such that $v' \in L(v_1, v_2, \dots, v_p)$. Then the set $\{v_1, v_2, \dots, v_p, v'\}$ is also linearly independent subset of

Let $S = \{0 = u_1, u_2, \dots, u_n\}$ be a set consisting of the zero vector. Then for any $y' = 0, y u_1 + 0 u_2 +$

$\dots + 0 u_n = 0$. Hence, for the system $\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n = 0$, we have a non-zero solution $\alpha_i = y$

and $\alpha = \alpha_2 = \dots = \alpha_n$. Therefore, the set S is linearly dependent. Theorem 3.2.4 Let $\{v_1, v_2, \dots, v_p\}$ be a linearly independent subset of a vector space V . Suppose there

$\dots, v_p, v_{p+1}\}$ is linearly dependent, then v_{p+1} is a linear combination of $v_1, v_2, \dots, v_p$. Proof. Since the set $\{v_1, v_2, \dots, v_p, v_{p+1}\}$ is linearly dependent, there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_{p+1}$ not all zero such that $\alpha_1 v_1 + \alpha_2 v_2 + \dots$

$+ \alpha_p v_p + \alpha_{p+1} v_{p+1} = 0$. (3.2.2)

Claim: $\alpha_{p+1} \neq 0$. Let if possible $\alpha_{p+1} = 0$. Then equation (3.2.2) gives $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_p v_p = 0$ with not all $\alpha_i, 1 \leq i \leq p$ zero. Hence, by the definition of linear independence, the set $\{v_1, v_2, \dots, v_p\}$ is linearly dependent

which is contradictory to our hypothesis. Thus, $\alpha_{p+1} \neq 0$ and we get $v_{p+1} = -\frac{1}{\alpha_{p+1}}(\alpha_1 v_1 + \dots + \alpha_p v_p)$.

8. Show that any set of k vectors in $\mathbb{R}^3$ is linearly dependent if $k \geq 4$. 9. Is the set of vectors $(1, 0), (i, 0)$ linearly independent subset of $C^2(\mathbb{R})$? 10. Under what conditions on α are the vectors $(1 + \alpha, 1 - \alpha)$ and $(\alpha - 1, 1 + \alpha)$

in $C^2(\mathbb{R})$ linearly independent? 11. Let $u, v \in V$ and M be a subspace of V . Further, let K be the subspace spanned by M and u and H be the subspace spanned by M and v . Show that if $v \in K$ and $v' \in M$ then $u \in H$. 3.3 Bases

Definition 3.3.1 (Basis of a Vector Space) 1. A non-empty subset B of a vector space V is called a basis of V if

$L(S)$?

(a) B is a linearly independent set, and

(b) $L(B) = V$, i.e., every vector in V can be expressed as a linear combination of the elements of B . independent?

space over $\mathbb{R}$. 8. Recall that $M_n(\mathbb{R})$ is the real vector space of all $n \times n$ real matrices. Prove that the following subsets are

subspaces of $M_n(\mathbb{R})$. (a) $sl_n = \{A \in M_n(\mathbb{R}) : \text{trace}(A) = 0\}$

(b) $\text{Sym}_n = \{A \in M_n(\mathbb{R}) : A = A^t\}$

(c) $\text{Skew}_n = \{A \in M_n(\mathbb{R}) : A + A^t = 0\}$

9. Let $V = \mathbb{R}$. Define $x \oplus y = x - y$ and $\alpha x = -\alpha x$. Which vector space axioms are not satisfied here? In this section, we saw

that a vector space has infinite number of vectors. Hence, one can start with any finite collection of vectors and obtain their

span. It means that any vector space contains infinite number of other vector subspaces. Therefore, the following questions

arise: 1. What are the conditions under which, the linear span of two distinct sets the same? 2. Is it possible to find/choose

vectors so that the linear span of the chosen vectors is the whole vector space itself? 3. Suppose we are able to choose

certain vectors whose linear span is the whole space. Can we find the minimum number of such vectors? We try to answer

these questions in the subsequent sections. 58 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES Note that $\alpha_i \in \mathbb{F}$

for every $i, 1 \leq i \leq p + 1$ and hence $-\alpha_{\alpha_{p+1}} \in \mathbb{F}$ for $1 \leq i \leq p$. Hence the result follows. We now state two important

corollaries of the above theorem. We don't give their proofs as they are easy consequence of the above theorem. Corollary

3.2.5 Let $\{u_1, u_2, \dots, u_n\}$ be a linearly dependent subset of a vector space V . Then there exists a smallest $k, 2 \leq k \leq n$ such that

V . Exercise 3.2.7 1. Consider the vector space $\mathbb{R}^2$. Let $u_1 = (1, 0)$. Find all choices for the vector u_2 such that the set $\{u_1,$

$u_2\}$ is linearly independent subset of $\mathbb{R}^2$. Does there exist choices for vectors u_2 and u_3 such that the set $\{u_1, u_2, u_3\}$ is

linearly independent subset of $\mathbb{R}^2$? 2. If none of the elements appearing along the principal diagonal of a lower triangular

matrix is zero, show that the row vectors are linearly independent in $\mathbb{R}^n$. The same is true for column vectors. 3. Let $S =$

$\{(1, 1, 1, 1), (1, -1, 1, 2), (1, 1, -1, 1)\} \subset \mathbb{R}^4$. Determine whether or not the vector $(1, 1, 2, 1) \in 4$. Show that $S = \{(1, 2,$

$3), (-2, 1, 1), (8, 6, 10)\}$ is linearly dependent in $\mathbb{R}^3$. 5. Show that $S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a linearly independent

set in $\mathbb{R}^3$. In general if $\{f_1, f_2, f_3\}$ is a linearly independent set then $\{f_1, f_1 + f_2, f_1 + f_2 + f_3\}$ is also a linearly independent set.

6. In $\mathbb{R}^3$, give an example of 3 vectors u, v and w such that $\{u, v, w\}$ is linearly dependent but any set of 2 vectors from u, v, w is linearly independent.

7. What is the maximum number of linearly independent vectors in $\mathbb{R}^3$?

3.3. BASES 59 2. A vector in B is called a basis vector. Remark 3.3.2 Let $\{v_1, v_2, \dots, v_p\}$ be a basis of a vector space V (F). Then any $v \in V$ is a unique

60 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES Remark 3.3.5 We can use the above results to obtain a basis of any finite dimensional vector space V as follows: Step 1: Choose a non-zero vector, say, $v_1 \in V$. Then the set $\{v_1\}$ is linearly independent. Step 2: If $V = L(v_1)$, we have got a basis of V . Else there exists a vector, say, $v_2 \in V$ such that

linear combination of the basis vectors, $v_1, v_2, \dots, v_p$. Observe that if there exists a $v \in W$ such that $v = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_p v_p$ and $v = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_p v_p$ then

$v_2 \in L(v_1)$. Then by Corollary 3.2.6, the set $\{v_1, v_2\}$ is linearly independent. Step 3: If $V = L(v_1, v_2)$, then $\{v_1, v_2\}$ is a basis of V . Else there exists a vector, say, $v_3 \in V$ such $0 = v - v = (\alpha_1 - \beta_1)v_1 + (\alpha_2 - \beta_2)v_2 + \dots + (\alpha_p - \beta_p)v_p$. But then the set $\{v_1, v_2, \dots, v_p\}$ is linearly independent and therefore the scalars $\alpha_i - \beta_i$ for $1 \leq i \leq p$ must all be equal to zero. Hence, for $1 \leq i \leq p$, $\alpha_i = \beta_i$ and we have the uniqueness. By convention, the linear span of an empty set is $\{0\}$. Hence, the empty set is a basis of the vector

space $\{0\}$. Example 3.3.3 1. Check that if $V = \{(x, y, 0) : x, y \in \mathbb{R}\} \subset \mathbb{R}^3$, then $B = \{(1, 0, 0), (0, 1, 0)\}$ or $B = \{(1, 0, 0), (1, 1, 0)\}$ or $B = \{(2, 0, 0), (1, 3, 0)\}$ or $\dots$ are bases of V .

2. For $1 \leq i \leq n$, let $e_i = (0, \dots, 0, 1$

th place, $0, \dots, 0) \in \mathbb{R}^n$. Then, the set $B = \{e_1, e_2, \dots, e_n\}$ forms a basis of $\mathbb{R}^n$. This set is called the standard basis of $\mathbb{R}^n$. That is, if $n = 3$, then the set $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ forms an standard basis of $\mathbb{R}^3$. that $v_3 \in L(v_1, v_2)$. So, by Corollary 3.2.6, the set $\{v_1, v_2, v_3\}$ is linearly independent. At the i th step, either $V = L(v_1, v_2, \dots, v_i)$, or $L(v_1, v_2, \dots, v_i) \neq V$. In the first case, we have $\{v_1, v_2, \dots, v_i\}$ as a basis of V . In the

second case, $L(v_1, v_2, \dots, v_i) \subset V$. So, we choose a vector, say, $v_{i+1} \in V$ such that $v_{i+1} \notin L(v_1, v_2, \dots, v_i)$. Therefore, by Corollary 3.2.6, the set $\{v_1, v_2, \dots, v_{i+1}\}$ is linearly independent. This process will finally end as V is

a finite dimensional vector space. Exercise 3.3.6 1. Let $S = \{v_1, v_2, \dots, v_p\}$ be a subset of a vector space V (F). Suppose $L(S) = V$ but S is not a linearly independent set. Then prove that each vector in V can be expressed

in more than one way as a linear combination of vectors from S . 2. Show that the set $\{(1, 0, 1), (1, i, 0), (1, 1, 1 - i)\}$ is a basis of $C^2(\mathbb{C})$.

3. Let $V = \{(x, y, z) : x+y+z = 0, x, y, z \in \mathbb{R}\}$ be a vector subspace of $\mathbb{R}^3$. Then $S = \{(1, 1, 2), (2, 1, 3), (1, 2, 3)\} \subset V$. It can be easily verified that the vector $(3, 2, 5) \in V$ and

$(3, 2, 5) = (1, 1, 2) + (2, 1, 3) = 4(1, 1, 2) - (1, 2, 3)$. Then by Remark 3.3.2, S cannot be a basis of V . A basis of V of z with $x + y$ to get A are linearly independent and they form a basis of the row space of A .

can be obtained by the following method: The condition $x + y + z = 0$ is equivalent to $z = -x - y$, we replace the value $(x, y, z) = (x, y, x + y) = (x, 0, x) + (0, y, y) = x(1, 0, 1) + y(0, 1, 1)$. Hence, $\{(1, 0, 1), (0, 1, 1)\}$ forms a basis of V .

4. Let $V = \{a + ib : a, b \in \mathbb{R}\}$ and $F = \mathbb{C}$. That is, V is a complex vector space. Note that any element $a + ib \in V$ can be written as $a + ib = (a + ib)1$. Hence, a basis of V is $\{1\}$.

3.3.1 Important Results

Theorem 3.3.7 Let $\{v_1, v_2, \dots, v_n\}$ be a basis of a given vector space V . If $\{w_1, w_2, \dots, w_m\}$ is a set of vectors from V with $m > n$ then this set is linearly dependent. Proof. Since we want to find whether the set $\{w_1, w_2, \dots$

5. Let $V = \{a + ib : a, b \in \mathbb{R}\}$ and $F = \mathbb{R}$. That is, V is a real vector space. Any element $a + ib \in V$ is expressible as $a \cdot 1 + b \cdot i$. Hence a basis of V is $\{1, i\}$.

Observe that i is a vector in $\mathbb{C}$. Also, $i' \in \mathbb{R}$ and hence $i \cdot (1 + 0 \cdot i)$ is not defined. $n, 1 \leq j \leq m$, such that

$w_m = a_{1m}v_1 + a_{2m}v_2 + \dots + a_{nm}v_n$. The set of equations (3.3.1) can be rewritten as $\square + \dots + a_m \square = 0$

62 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES Note that the Corollary 3.2.6 can be used to generate a basis of $v_2 + \dots + v_m$

(Finite Dimensional Vector Space) A vector space V is said to be finite dimensional if there exists a basis consisting of finite number of

elements. Otherwise, the vector space V is called infinite dimensional. In Example 3.3.3, the vector space of all polynomials is an

example of an infinite dimensional vector space. All the other vector spaces are finite dimensional. 3.3. BASES 61 Since the set $\{v_1, v_2,$

$\dots, v_n\}$ is linearly independent, we have $\sum_{i=1}^n \alpha_i a_{ni} = 0$. Therefore, finding α_i 's satisfying equation (3.3.1) reduces to solving the system of

homogeneous equations $\sum_{i=1}^n \alpha_i a_{ni} = \dots = \alpha_m$

$A\alpha = 0$ where $\alpha^t = (\alpha_1, \alpha_2, \dots, \alpha_m)$ and $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \end{bmatrix}$

$$w_1 = a_{11}v_1 + a_{21}v_2 + \dots + a_{n1}v_n$$

$$w_2 = a_{12}v_1 + a_{22}v_2 + \dots + a_{n2}v_n$$

$(a + ib, c + id) = a(1, 0) + b(i, 0) + c(0, 1) + d(0, i)$. Hence, the set $\{(1, 0), (i, 0), (0, 1), (0, i)\}$ is a basis and $\dim(V) = 4$. Remark 3.3.13 It is important to note that the dimension of a vector space may change if the underlying field (the set of scalars) is changed. Example

3.3.14 Let V be the set of all functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with the property that $f(x+y) = f(x) + f(y)$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\sum_{j=1}^n a_{1j} v_j = \sum_{j=1}^n a_{2j} v_j$$

$$\text{i.e., } \sum_{j=1}^n a_{1j} v_j + \sum_{j=1}^n a_{2j} v_j$$

set consists of infinite number of elements. Therefore, the equation (3.3.1) has a dependent set. Remark 3.3.8 Let V be a vector subspace of $\mathbb{R}^n$ with spanning set S .

solution with not all $\alpha_i, 1 \leq i \leq m$, zero. Hence, the set $\{w_1, w_2, \dots, w_m\}$ is a linearly We give a method of finding a basis of V from S .

2. Use only the elementary row operations $R_i(c)$ and $R_j(c)$ to get the row-reduced form B of A (in fact we just need to make as

many zero-rows as possible). 3. Let B be the set of vectors in S corresponding to the non-zero rows of B . Then the set B is a

basis of $L(S) = V$. Example 3.3.9 Let $S = \{(1, 1, 1, 1), (1, 1, -1, 1), (1, 1, 0, 1), (1, -1, 1, 1)\}$ be a subset of $\mathbb{R}^4$. Find a basis of $(f \circ g)(x) = f(g(x))$ and $(t \circ f)(x) = f(tx)$.

and $f(ax) = af(x)$. For $f, g \in V$, and $t \in \mathbb{R}$, define

$$\sum_{i=1}^m \alpha_i a_{i1} = \sum_{i=1}^m \alpha_i$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix}$$

1 - 1 1 1 Applying row-reduction to A , we have $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_{12}(-1), R_{13}(-1), R_{14}(-1)} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_{32}(-2)} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Then V is a real vector space. For $1 \leq i \leq n$, consider the functions

$e_i(x) = e_i(x_1, x_2, \dots, x_n) = x_i$. Then it can be easily verified that the set $\{e_1, e_2, \dots, e_n\}$ is a basis of V and hence $\dim(V) = n$. The next theorem follows directly from Corollary 3.2.6 and Theorem 3.3.7. Hence, the proof

is omitted. Theorem 3.3.15 Let S be a linearly independent subset of a finite dimensional vector space V . Then the set S can be extended to form a basis of V . Theorem 3.3.15 is equivalent to the following statement:

Let V be a vector space of dimension n . Suppose, we have found a linearly independent set $S =$

$\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 \end{pmatrix}$

Observe that the rows 1, 3 and 4 are non-zero. Hence, a basis of $L(S)$ consists of the first, third and fourth vectors of the set S . Thus, $B = \{(1, 1, 1, 1), (1, 1, 0, 1), (1, -1, 1, 1)\}$ is a basis of $L(S)$. Observe

that at the last step, in place of the elementary row operation $R_{32}(-2)$, we can apply $R_{23}(-\frac{1}{2})$ to make the third row as the zero-row. In this case, we get $\{(1, 1, 1, 1), (1, 1, -1, 1), (1, -1, 1, 1)\}$ as a basis of $L(S)$. Corollary

3.3.10 Let V be a finite dimensional vector space. Then any two bases of V have the same number of vectors. Proof. Let $\{u_1, u_2, \dots, u_m\}$ and $\{v_1, v_2, \dots, v_n\}$ be two bases of V with $m > n$. Then by the above

$\{v_1, v_2, \dots, v_n\} \subset V$. Then there exist vectors $v_{n+1}, \dots, v_m$ in V such that $\{v_1, v_2, \dots, v_m\}$ is a basis of V . Corollary 3.3.16 Let V be a vector space of dimension n . Then any set of n linearly independent vectors forms a basis

of V . Also, every set of m vectors, $m > n$, is linearly dependent. Example 3.3.17 Let $V = \{(v, w, x, y, z) \in \mathbb{R}^5 : v + x - 3y + z = 0\}$ and $W = \{(v, w, x, y, z) \in \mathbb{R}^5 : w - x - z = 0, v = y\}$ be two subspaces of $\mathbb{R}^5$. Find bases of V

and W containing a basis of $V \cap W$. Solution: Let us find a basis of $V \cap W$. The solution set of the linear equations

$v + x - 3y + z = 0, w - x - z = 0$ and $v = y$ is given by

theorem the set $\{v_1, v_2, \dots, v_m\}$ is linearly dependent if we take $\{u_1, u_2, \dots, u_n\}$ as the basis of V . This contradicts the assumption that $\{v_1, v_2, \dots, v_m\}$ is also a basis of V . Hence, we get $m = n$. Definition 3.3.11 (Dimension of a Vector Space) The dimension of a finite dimensional vector space V is denoted by $\dim(V)$.

is the number of vectors in a basis of V , denoted $\dim(V)$.

Solution: Here $A =$

$\begin{pmatrix} 1 & 2 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$

$(v, w, x, y, z)^T = (y, 2y, x, y, 2y - x)^T = y(1, 2, 0, 1, 2)^T + x(0, 0, 1, 0, -1)^T$. Thus, a basis of $V \cap W$ is $\{(1, 2, 0, 1, 2), (0, 0, 1, 0, -1)\}$. To find a basis of W containing a basis of $V \cap W$, we can proceed as follows:

3.3. BASES 63 1. Find a basis of W .

2. Take the basis of $V \cap W$ found above as the first two vectors and that of W as the next set of vectors. Now use Remark 3.3.8 to get the required basis. Heuristically, we can also find the basis in the following way: A

vector of W has the form $(y, x + z, x, y, z)$ for $x, y, z \in \mathbb{R}$. Substituting $y = 1, x = 1$, and $z = 0$ in

64 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES 7. If M and N are 4-dimensional subspaces of a vector space V of dimension 7 then show that M and N have at least one vector in common other than the zero

vector. 8. Let $P = L(\{(1, 0, 0), (1, 1, 0)\})$ and $Q = L(\{(1, 1, 1)\})$ be vector subspaces of $\mathbb{R}^3$. Show that $P + Q = \mathbb{R}^3$ and $P \cap Q = \{0\}$. If $u \in \mathbb{R}^3$, determine u_P, u_Q such that $u = u_P + u_Q$ where $u_P \in P$ and $u_Q \in Q$. Is it necessary that u_P

and u_Q are unique?

9. Let W_1 be a k -dimensional subspace of an n -dimensional vector space V (F) where $k \geq 1$. Prove that there exists an $(n - k)$ -dimensional subspace W_2 of V such that $W_1 \cap W_2 = \{0\}$ and $W_1 + W_2 = V$.

($y, x + z, x, y, z$) gives us the vector $(1, 1, 1, 1, 0) \in W$. It can be easily verified that a basis of W is

10. Let P and Q be subspaces of $\mathbb{R}^3$ such that $P + Q = \mathbb{R}^3$ and $P \cap Q = \{0\}$. Then show that each $\{(1, 2, 0, 1, 2), (0, 0, 1, 0, -1), (1, 1, 1, 1, 0)\}$.

Similarly, a vector of V has the form $(v, w, x, y, 3y - v - x)$ for $v, w, x, y \in \mathbb{R}$. Substituting $v = 0, w = 1, x = 0$ and $y = 0$, gives a vector $(0, 1, 0, 0, 0) \in V$. Also, substituting $v = 0,$

$w = 1, x = 1$ and $y = 1$, gives another vector $(0, 1, 1, 1, 2) \in V$. So, a basis of V can be taken as

$u \in \mathbb{R}^3$ can be uniquely expressed as $u = u_P + u_Q$ where $u_P \in P$ and $u_Q \in Q$. 11. Let $P = L(\{(1, -1, 0), (1, 1, 0)\})$ and $Q = L(\{(1, 1, 1), (1, 2, 1)\})$ be vector subspaces of $\mathbb{R}^3$. Show that $P + Q = \mathbb{R}^3$ and $P \cap Q = \{0\}$. Show that

there exists a vector $u \in \mathbb{R}^3$ such that u cannot be written uniquely in the form $u = u_P + u_Q$ where $u_P \in P$ and $u_Q \in Q$. 12. Recall the vector space $P_n(\mathbb{R})$. Is the set,

$\{(1, 2, 0, 1, 2), (0, 0, 1, 0, -1), (0, 1, 0, 0, 0), (0, 1, 1, 1, 2)\}$ Recall that for two vector subspaces M and N of a vector space V (F), the vector subspace $M + N$ is defined by

$W = \{p(x) \in P_n(\mathbb{R}) : p(-1) = p(1) = 0\}$

$M + N = \{u + v : u \in M, v \in N\}$. With this definition, we have the following very important theorem (for a proof, see Appendix 14.4.1).

for different values of m integrate the function

$\int_0^1 x^m dx$

Theorem 3.3.18 Let V (F) be a finite dimensional vector space and let M and N be two subspaces of V . $\dim(M) + \dim(N) = \dim(M + N) + \dim(M \cap N)$.

(b) the matrices P_1 and P_2 such that $P_1 A$ and $P_2 B$ are in row-reduced form. (c) a basis each for the row spaces of A and B . (d) a basis

+ $\dim(M \cap N)$. (3.3.2) Exercise 3.3.19 1. Find a basis of the vector space $P_n(\mathbb{R})$. Also, find $\dim(P_n(\mathbb{R}))$. What can you say about the

each for the range spaces of A and B . (e) bases of the null spaces of A and B .

$\int_0^1 \sin(mx) (\alpha_1 \sin(k_1 x) + \alpha_2 \sin(k_2 x) + \dots + \alpha_n \sin(k_n x)) dx$ to get the required result.]

dimension of $P(\mathbb{R})$?

a subspace of $P_4(\mathbb{R})$? If yes, find its dimension. 13. Let V be the set of all 2×2 matrices with complex entries and $a_{11} + a_{22}$

3. Show that the set $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis of $C^3(\mathbb{C})$. Is it a basis of $C^3(\mathbb{R})$ also?

(f) the dimensions of all the vector subspaces so obtained. 15. Let $M(n, \mathbb{R})$ denote the space of all $n \times n$ real matrices.

= 0. Show that V is a real vector space. Find its basis. Also let $W = \{A \in V : a_{21} = -a_{12}\}$. Show W is a vector subspace of V ,

For the sets given below, check that they are subspaces of $M(n, \mathbb{R})$ and also find their dimension. (a) $sl(n, \mathbb{R}) = \{A$

and find its dimension. 14. Let $A = \begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 0 & 6 & -1 \\ 0 & -1 & 1 & 2 & 0 \end{pmatrix}$

$\in M(n, \mathbb{R}) : \text{tr}(A) = 0\}$, where recall that $\text{tr}(A)$ stands for trace of A . (b) $S(n, \mathbb{R}) = \{A \in M(n, \mathbb{R}) : A = A^T\}$.

$\begin{pmatrix} 4 & 2 & 5 & 6 & 10 \\ 0 & 1 & 1 & 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 4 & 0 & 6 & -1 \\ 0 & -1 & 1 & 2 & 0 \end{pmatrix}$ be two matrices. For

4. Let $W = \{(x, y, z, w) \in \mathbb{R}^4 : x + y - z + w = 0\}$ be a subspace of $\mathbb{R}^4$. Find its basis and dimension. 5. Let $V = \{(x, y, z, w) \in \mathbb{R}^4 : x + y - z + w = 0, x + y + z + w = 0\}$ and $W = \{(x, y, z, w) \in \mathbb{R}^4 :$

Then

$x - y - z + w = 0, x + 2y - w = 0\}$ be two subspaces of $\mathbb{R}^4$. Find bases and dimensions of V, W . 6. Let V be the set of all real

symmetric $n \times n$ matrices. Find its basis and dimension. What if V is the complex vector space of all $n \times n$ Hermitian matrices?

(c) $A(n, \mathbb{R}) = \{A \in M(n, \mathbb{R}) : A + A^T = 0\}$. Before going to the next section, we prove that for any matrix A of order $m \times n$ Row rank(A) = Column rank(A).

= $\sin(nx)$. Prove that the collection of vectors $\{e_n : 1 \leq n < \infty\}$ is a linearly independent set.

[Hint: On the contrary, assume that the set is linearly dependent. Then we have a finite set of vectors, say $\{e_{n_1}, e_{n_2}, \dots, e_{n_r}\}$ that are linearly

dependent. That is, there exist scalars $\alpha_i \in \mathbb{R}$ for $1 \leq i \leq r$ not all zero such that $\alpha_1 \sin(k_1 x) + \alpha_2 \sin(k_2 x) + \dots + \alpha_r \sin(k_r x) = 0$ for all $x \in [0, 2\pi]$. Now

3.3. BASES 65 Proposition 3.3.20 Let A be an $m \times n$ real matrix. Then Row rank(A) = Column rank(A).

0^{2n}

Proof. Let $R_1, R_2, \dots, R_m$ be the rows of A and $C_1, C_2, \dots, C_n$ be the columns of A . Note that Row rank(A) = r , means that $\dim L(R_1, R_2, \dots, R_m) = r$. Hence, there exists vectors $u_1 = (u_{11}, \dots, u_{1n}), u_2 = (u_{21}, \dots, u_{2n}), \dots$

$\dots, u_r = (u_{r1}, \dots, u_{rn}) \in \mathbb{R}^n$

$V \cap W$ and $V + W$.

$R_i \in L(u_1, u_2, \dots, u_r) \in \mathbb{R}^n$ for all $i, 1 \leq i \leq m$. Therefore, there exist real numbers $\alpha_{ij}, 1 \leq i \leq m, 1 \leq j \leq r$ such that $R_i = \alpha_{i1}u_1 + \alpha_{i2}u_2 + \dots + \alpha_{ir}u_r =$

$\sum_{j=1}^r \alpha_{ij}u_j, \dots,$

$R_2 = \alpha_{21}u_1 + \alpha_{22}u_2 + \dots + \alpha_{2r}u_r =$

$\sum_{j=1}^r \alpha_{2j}u_j, \dots,$

$\sum_{j=1}^r \alpha_{2j}u_j$ and so on, till $R_m = \alpha_{m1}u_1 + \dots + \alpha_{mr}u_r =$

$$\alpha_{m1}u_{12}, \dots, \alpha_{mr}$$

$$\alpha_{11}u_{11} + \alpha_{12}u_{12} + \dots + \alpha_{1r}u_{1r}$$

$$\alpha_{21}u_{11} + \alpha_{22}u_{12} + \dots + \alpha_{2r}u_{1r}$$

$$\vdots$$

$$\alpha_{m1}u_{11} + \alpha_{m2}u_{12} + \dots + \alpha_{mr}u_{1r}$$

In general, for $1 \leq j \leq n$, we have with

$$\alpha_{11}u_{11} + \alpha_{12}u_{12} + \dots + \alpha_{1r}u_{1r}$$

$$\vdots$$

$$\alpha_{m1}u_{11} + \alpha_{m2}u_{12} + \dots + \alpha_{mr}u_{1r}$$

Therefore, we observe that the columns $C_1, C_2, \dots, C_n$ are linear combination of the r vectors $(\alpha_{11}, \alpha_{21}, \dots, \alpha_{m1})^T, (\alpha_{12}, \alpha_{22}, \dots, \alpha_{m2})^T, \dots, (\alpha_{1r}, \alpha_{2r}, \dots, \alpha_{mr})^T$.

$$\alpha_{11}u_{11} + \alpha_{12}u_{12} + \dots + \alpha_{1r}u_{1r}$$

$$\vdots$$

$$\alpha_{m1}u_{11} + \alpha_{m2}u_{12} + \dots + \alpha_{mr}u_{1r}$$

Therefore, $\text{Column rank}(A) = \dim L(C_1, C_2, \dots, C_n) \leq r = \text{Row rank}(A)$. A similar argument gives $\text{Row rank}(A) \leq \text{Column rank}(A)$. Thus, we have the required result.

66 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES 3.4 Ordered Bases

Let $B = \{u_1, u_2, \dots, u_n\}$ be a basis of a vector space $V(F)$. As B is a set, there is no order of its elements. In this section, we want to associate an order among the vectors in any basis of V . **Definition 3.4.1 (Ordered Basis)** An ordered basis for a vector space $V(F)$ of dimension n , is a basis $\{u_1, u_2, \dots, u_n\}$ together with a one-to-one correspondence between the sets $\{u_1, u_2, \dots, u_n\}$ and $\{1, 2, 3, \dots, n\}$. If the ordered basis has u_1 as the first vector, u_2 as the second vector and so on, then we denote this ordered basis by $(u_1, u_2, \dots, u_n)$. **Example 3.4.2** Consider $P_2(\mathbb{R})$, the vector space of all polynomials of degree less than or equal to 2 with coefficients from $\mathbb{R}$. The set $\{1 - x, 1 + x, x^2\}$ is a basis of $P_2(\mathbb{R})$. For any element $a_0 + a_1x + a_2x^2 \in P_2(\mathbb{R})$, we have $2(1 - x) + a_0 + a_1$

$$2(1 - x) + a_2x^2$$

$$2$$

is the second component, and a_2 is the third component of the vector $a_0 + a_1x + a_2x^2$. If $(1 - x, 1 + x, x^2)$ is an ordered basis, then $a_0 - a_1$ is the first component, $a_0 + a_1$

is the first component, $a_0 - a_1$ is the second component, and a_2 is the third component of the vector $a_0 + a_1x + a_2x^2$. That is, as ordered bases $(u_1, u_2, \dots, u_n)$, $(u_2, u_3, \dots, u_n, u_1)$, and $(u_n, u_1, u_2, \dots, u_{n-1})$ are different even though they have the same set of vectors as elements.

Definition 3.4.3 (Coordinates of a Vector) Let $B = (v_1, v_2, \dots, v_n)$ be an ordered basis of a vector space V . If we take $(1 + x, 1 - x, x^2)$ as an ordered basis, then $a_0 + a_1x \in V(F)$ and let $v \in V$. If

$$v = \beta_1v_1 + \beta_2v_2 + \dots + \beta_nv_n$$

then the tuple $(\beta_1, \beta_2, \dots, \beta_n)$ is called the coordinate of the vector v with respect to the ordered basis B . Mathematically, we denote it by $[v]_B = (\beta_1, \dots, \beta_n)^T$, a column vector. Suppose $B_1 = (u_1, u_2, \dots, u_n)$ and $B_2 = (u_n, u_1, u_2, \dots, u_{n-1})$ are two ordered bases of V . Then for any $x \in V$ there exists unique scalars $\alpha_1, \alpha_2, \dots, \alpha_n$ such that

$$x = \alpha_1u_1 + \alpha_2u_2 + \dots + \alpha_nu_n = \alpha_nu_n + \alpha_1u_1 + \dots + \alpha_{n-1}u_{n-1}$$

$$[x]_{B_1} = (\alpha_1, \alpha_2, \dots, \alpha_n)^T \text{ and } [x]_{B_2} = (\alpha_n, \alpha_1, \alpha_2, \dots, \alpha_{n-1})^T$$

Note that x is uniquely written as I^n

$$a_0 + a_1x + a_2x^2 = a_0 - a_1$$

$$a_{u_1}$$

and hence the coordinates with respect to an ordered basis are unique. Suppose that the ordered basis B_1 is changed to the ordered basis $B_2 = (u_2, u_1, u_3, \dots, u_n)$. Then $[x]_{B_2} = (\alpha_2, \alpha_1, \alpha_3, \dots, \alpha_n)^T$. So, the coordinates of a vector depend on the ordered basis chosen. **Example 3.4.4** Let $V = \mathbb{R}^3$. Consider the ordered bases V . Then, with respect to the above bases we have

So,

$$I^T$$

$$\alpha_{11}$$

$$\alpha_{12}$$

$$\vdots$$

$$\alpha_{m1}$$

$$\alpha_{m2}$$

$$\vdots$$

$$\alpha_{mr}$$

$$(1, -1, 1) = 1 \cdot (1, 0, 0) + (-1) \cdot (0, 1, 0) + 1 \cdot (0, 0, 1) = 2 \cdot (1, 0, 0) + (-2) \cdot (1, 1, 0) + 1 \cdot (1, 1, 1)$$

$$\alpha_{11}u_{11} + \alpha_{12}u_{12} + \dots + \alpha_{1r}u_{1r}$$

$$\vdots$$

$$\alpha_{m1}u_{11} + \alpha_{m2}u_{12} + \dots + \alpha_{mr}u_{1r}$$

$$C_1 = \alpha_{11}u_{11} + \alpha_{12}u_{12} + \dots + \alpha_{1r}u_{1r}$$

$$\vdots$$

$$\alpha_{m1}u_{11} + \alpha_{m2}u_{12} + \dots + \alpha_{mr}u_{1r}$$

$$\alpha_{11}u_{11} + \alpha_{12}u_{12} + \dots + \alpha_{1r}u_{1r}$$

$$\vdots$$

$$\alpha_{m1}u_{11} + \alpha_{m2}u_{12} + \dots + \alpha_{mr}u_{1r}$$

$$C_j = \sum_{i=1}^n \alpha_{ij} u_i$$

$$= u_{1j} \square + u_{2j} \square$$

Therefore,

21.
22.

$$C_j = \sum_{i=1}^n \alpha_{ij} u_i$$

and $B_3 = (1, 1, 1), (1, 1, 0), (1, 0, 0)$ of

3.4. ORDERED BASES 67 Therefore, if we write $u = (1, -1, 1)$, then $[u]_{B_1} = (1, -1, 1)^t$, $[u]_{B_2} = (2, -2, 1)^t$, $[u]_{B_3} = (1, -2, 2)^t$. In general, let V be an n -dimensional vector space with ordered bases $B_1 = (u_1, u_2, \dots, u_n)$ and

$B_2 = (v_1, v_2, \dots, v_n)$. Since, B_1 is a basis of V , there exists unique scalars a_{ij} , $1 \leq i, j \leq n$ such that $a_{ij}u_i$ for $1 \leq i \leq n$. That is, for each i , $1 \leq i \leq n$, $[v_i]_{B_1} = (a_{i1}, a_{i2}, \dots, a_{in})^t$.

Let $v \in V$ with $[v]_{B_1} = (\alpha_1, \alpha_2, \dots, \alpha_n)^t$. As B_2 as ordered basis $(v_1, v_2, \dots, v_n)$, we have Since B_1 is a basis this representation of v in terms of u_i 's is unique. So,

$$v = \alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n$$

$= A[v]_{B_2}$. Note that the i^{th} column of the matrix A is equal to $[v_i]_{B_1}$, i.e., the i^{th} column of A is the coordinate of the i^{th} vector v_i of B_2 with respect to the ordered basis B_1 . Hence, we have proved the following

theorem. Theorem 3.4.5 Let V be an n -dimensional vector space with ordered bases $B_1 = (u_1, u_2, \dots, u_n)$ and $B_2 = (v_1, v_2, \dots, v_n)$. Let

$$A = [[v_1]_{B_1}, [v_2]_{B_1}, \dots, [v_n]_{B_1}]$$

$$v = \sum_{i=1}^n \alpha_i v_i$$

68 CHAPTER 3. FINITE DIMENSIONAL VECTOR SPACES2. Let $A = [a_{ij}] = \begin{bmatrix} \square & \square & \square \\ 0 & 2 & 0 \end{bmatrix}$

1 1 0. The columns of the matrix A are obtained by the following rule: $0 - 2 \ 1$

$[(1, 1, 1)]_{B_1} = 0 \cdot (1, 0, 0) + 0 \cdot (1, 1, 0) + 1 \cdot (1, 1, 1) = (0, 0, 1)^t$,
 $[(1, -1, 1)]_{B_1} = 2 \cdot (1, 0, 0) + (-2) \cdot (1, 1, 0) + 1 \cdot (1, 1, 1) = (2, -2, 1)^t$ $[(1, 1, 0)]_{B_1} = 0 \cdot (1, 0, 0) + 1 \cdot (1, 1, 0) + 0 \cdot (1, 1, 1) = (0, 1, 0)^t$ That is, the elements of $B_2 = (1, 1, 1), (1, -1, 1), (1, 1, 0)$ are expressed in terms of the ordered basis3. Note that for any $(x, y, z) \in \mathbb{R}^3$,

$$[(x, y, z)]_{B_1} = \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

4. The matrix A is invertible and hence $[(x, y, z)]_{B_2} = A^{-1}[(x, y, z)]_{B_1}$. In the next chapter, we try to understand Theorem 3.4.5 again using the ideas of 'linear transformations / functions'. Exercise 3.4.7 1. Determine the coordinates of the vectors $(1, 2, 1)$ and $(4, -2, 2)$ with respect to the2. Consider the vector space $P_3(\mathbb{R})$.

(a) Show that $B_1 = (1 - x, 1 + x^2, 1 - x^3, 3 + x^2 - x^3)$ and $B_2 = (1, 1 - x, 1 + x^2, 1 - x^3)$ are bases(b) Find the coordinates of the vector $u = 1 + x + x^2 + x^3$ with respect to the ordered basis B_1 and

and (c) Find the matrix A such that $[u]_{B_2} = A[u]_{B_1}$. (d) Let $v = a_0 + a_1x + a_2x^2 + a_3x^3$. Then verify the following:

$$-a_0 - a_1 + 2a_2 - a_3$$

$$[v]_{B_1} = A[v]_{B_2}$$

Example 3.4.6 Consider two bases $B_1 = (1, 0, 0), (1, 1, 0), (1, 1, 1)$ and $B_2 = (1, 1, 1), (1, -1, 1), (1, 1, 0)$

$$[v]_{B_1} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

$$[(x, y, z)]_{B_1} = (x - y) \cdot (1, 0, 0) + (y - z) \cdot (1, 1, 0) + z \cdot (1, 1, 1)$$

$$= (x - y, y - z, z)^t$$

Chapter 4

Linear Transformations

$$[(x, y, z)]_{B_2} = (y - z) \cdot (1, 1, 1) + (2 + z) \cdot (1, 1, 1) + \frac{x - y}{2} \cdot (1, -1, 1)$$

$$= (y - x, 2 + z, \frac{x - y}{2} \cdot x - z)^t$$

of $\mathbb{R}^3$.

basis $B = \{(2, 1, 0), (2, 1, 1), (2, 2, 1)\}$ of $\mathbb{R}^3$.

of $P_3(\mathbb{R})$.

B_2 .

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Proposition 4.1.3 Let $T : V \rightarrow W$ be a linear transformation. Suppose that 0_V is the zero vector in V and

$$-a_1$$

$$[v]_{B_1} =$$

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So, $T(0_V) = 0_W$ as $T(0_V) \in W$. From now on, we write 0 for both the zero vector of the domain space and the zero

vector of the range space. Definition 4.1.4 (Zero Transformation) Let V be a

vector space and let $T : V \rightarrow W$ be the map defined $T(v) = 0$ for every $v \in V$.

Then T is a linear transformation. Such a linear transformation is called the zero transformation and is denoted by 0. Definition 4.1.5

(Identity Transformation) Let V be a vector space and let $T : V \rightarrow V$ be the map defined by

4.1 Definitions and Basic Properties

Throughout this chapter, the scalar field F is either always the set $\mathbb{R}$ or always the set $\mathbb{C}$.

Definition 4.1.1 (Linear Transformation) Let V and W be vector

spaces over F . A map $T : V \rightarrow W$ is

$T(v) = v$ for every $v \in V$.

called a linear transformation if

$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$, for all $\alpha, \beta \in F$, and $u, v \in V$. We now give a few examples of linear transformations.

Example 4.1.2 1. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = (x, 3x)$ for all $x \in \mathbb{R}$. Then T is a linear transformation $T(x$

$+ y) = (x + y, 3(x + y)) = (x, 3x) + (y, 3y) = T(x) + T(y)$. 2. Verify that the maps given below from $\mathbb{R}^n$ to $\mathbb{R}$ are linear transformations. Let $x = (x_1, x_2, \dots, x_n)$. (a) Define $T(x) = I^n$

(b) For any i , $1 \leq i \leq n$, define $T_i(x) = x_i$.

(c) For a fixed vector $a = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, define $T(x) = I^n a x_i$. Note that examples (a)

and (b) can be obtained by assigning particular values for the vector a .

3. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T((x, y)) = (x + y, 2x - y, x + 3y)$. Then T is a linear transformation with $T((1, 0)) = (1, 2, 1)$ and $T((0, 1)) = (1, -1, 3)$.

4. Let A be an $m \times n$ real matrix. Define a map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T_A(x) = Ax$ for every $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Then T_A is a linear transformation. That is, every $m \times n$ real matrix defines a linear transformation 5. Recall that $P_n(\mathbb{R})$ is the set of all polynomials of degree less than or equal to n with real coefficients. $T((a_1, a_2, \dots,$

$a_{n+1})) = a_1 + a_2 x + \dots + a_{n+1} x^n$

as

for $(a_1, a_2, \dots, a_{n+1}) \in \mathbb{R}^{n+1}$. Then T is a linear transformation.

$\sum_{i=1}^n x_i$.

$\sum_{i=1}^n$

from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Define $T : \mathbb{R}^{n+1} \rightarrow P_n(\mathbb{R})$ by

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Then T is a linear transformation. Such a linear transformation is called the Identity transformation and is denoted by I .

We now prove a result that relates a linear transformation T with its value on a basis of the

domain. Theorem 4.1.6 Let $T : V \rightarrow W$ be a linear transformation and $B = (u_1, u_2, \dots, u_n)$ be an ordered basis of V . Then the linear transformation T is a linear combination of the vectors $T(u_1), T(u_2), \dots, T(u_n)$. In

other words, T is determined by $T(u_1), T(u_2), \dots, T(u_n)$.

Proof. Since B is a basis of V , for any $x \in V$, there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $x = \alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n$. So, by the definition of a linear transformation $T(x) = T(\alpha_1 u_1 + \dots + \alpha_n u_n) = \alpha_1 T(u_1) + \dots + \alpha_n T(u_n)$.

Observe that, given $x \in V$, we know the scalars $\alpha_1, \alpha_2, \dots, \alpha_n$. Therefore, to know $T(x)$, we just need to know the vectors $T(u_1), T(u_2), \dots, T(u_n)$ in W . That is, for every $x \in V$, $T(x)$ is determined by the coordinates $(\alpha_1,$

$\alpha_2, \dots, \alpha_n)$ of x with respect to the ordered basis B and the vectors $T(u_1), T(u_2), \dots, T(u_n) \in W$. Exercise 4.1.7 1. Which of the following are linear transformations $T : V \rightarrow W$? Justify your answers. (a) Let $V = \mathbb{R}^2$ and W

$= \mathbb{R}^3$ with $T(x, y) = (x + y + 1, 2x - y, x + 3y)$

by

(b) Let $V = W = \mathbb{R}^2$ with $T(x, y) = (x - y, x^2 - y^2)$ (c) Let $V = W = \mathbb{R}^2$ with $T(x, y) = (x - y, |x|)$ (d) Let $V = \mathbb{R}^2$ and $W = \mathbb{R}^4$ with $T(x, y) = (x + y, x - y, 2x + y, 3x - 4y)$ (e) Let $V = W = \mathbb{R}^4$ with $T(x, y, z, w) = (z, x, w, y)$

2. Recall that $M_2(\mathbb{R})$ is the space of all 2×2 matrices with real entries. Then, which of the following are linear transformations $T : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$?

4.1. DEFINITIONS AND BASIC PROPERTIES 71 (a) $T(A) = A^t(b)$ (b) $T(A) = I + A$ (c) $T(A) = A^2$

72 CHAPTER 4. LINEAR TRANSFORMATIONS Proof. Since T is onto, for each $w \in W$ there exists a vector $v \in V$ such that $T(v) = w$. So, the set $T^{-1}(w)$ is non-empty. Suppose there exist vectors $v_1, v_2 \in V$ such that T

$(v_1) = T(v_2)$. But by assumption, T is one-one and therefore $v_1 = v_2$. This completes the proof of Part 1. We now show that T^{-1} as defined above is a linear transformation. Let $w_1, w_2 \in W$. Then by Part 1, there exist unique

vectors $v_1, v_2 \in V$ such that $T^{-1}(w_1) = v_1$ and $T^{-1}(w_2) = v_2$. Or equivalently,

(d) $T(A) = BAB^{-1}$, where B is some fixed 2×2 matrix. 3. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ be a map. Then T is a linear transformation if and only if there exists a unique $c \in \mathbb{R}$ such that $T(x) = cx$ for every $x \in \mathbb{R}$.

4. Let A be an $n \times n$ real matrix. Consider the linear transformation

$T_A(x) = Ax$ for every $x \in \mathbb{R}^n$. Then prove that $T^2(x) := T(T(x)) = A^2 x$. In general, for $k \in \mathbb{N}$, prove that $T^k(x) = A^k x$. 5. Use the ideas of matrices to give examples of linear transformations $T, S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that satisfy: (a) $T^2 =$

$0, T^2 = 0, T^3 = 0$.

$$[T(x)]_{B_2} =$$

$$\begin{bmatrix} \square \\ \square \\ \square \\ \vdots \\ \square \end{bmatrix} =$$

$$\begin{bmatrix} \square \\ \square \\ \square \\ \vdots \\ \square \end{bmatrix}$$

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$$[T(x)]_{B_2} = A[x]_{B_1}$$

$$T((x, y)) = (x + y, x - y). \text{ We obtain } T[B_1, B_2], \text{ the matrix of the linear transformation } T \text{ with respect to the ordered bases } B_1 = (1, 0), (0, 1) \text{ and } B_2 = (1, 1), (1, -1) \text{ of } \mathbb{R}^2. \text{ For any vector } (x, y) \in \mathbb{R}^2, [(x, y)]_{B_1} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{as } (x, y) = x(1, 0) + y(0, 1). \text{ Also, by definition of the linear transformation } T, \text{ we have } T((1, 0)) = (1, 1) = 1 \cdot (1, 1) + 0 \cdot (1, -1). \text{ So, } [T((1, 0))]_{B_2} = (1, 0)^T. T((0, 1)) = (1, -1) = 0 \cdot (1, 1) + 1 \cdot (1, -1). \text{ That is, } [T((0, 1))]_{B_2} =$$

$$(0, 1)^T. \text{ So the } T[B_1, B_2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \text{ Observe that in this case,}$$

$$[T((x, y))]_{B_2} = [(x + y, x - y)]_{B_2} = x(1, 1) + y(1, -1) = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} x \\ -y \end{bmatrix} = \begin{bmatrix} x+x \\ y-y \end{bmatrix} = [T((x, y))]_{B_2}.$$

$$T[B_1, B_2] [(x, y)]_{B_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$2. \text{ Let } B_1 = (1, 0, 0), (0, 1, 0), (0, 0, 1), B_2 = (1, 0, 0), (1, 1, 0), (1, 1, 1) \text{ be two ordered bases of } \mathbb{R}^3. T: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \text{ by } T(x) = x.$$

$$T((1, 0, 0)) = 1 \cdot (1, 0, 0) + 0 \cdot (1, 1, 0) + 0 \cdot (1, 1, 1),$$

$$T((0, 1, 0)) = -1 \cdot (1, 0, 0) + 1 \cdot (1, 1, 0) + 0 \cdot (1, 1, 1), \text{ and } T((0, 0, 1)) = 0 \cdot (1, 0, 0) + (-1) \cdot (1, 1, 0) + 1 \cdot (1, 1, 1). \text{ Thus, we have } T[B_1, B_2] = [[T((1, 0, 0))]_{B_2}, [T((0, 1, 0))]_{B_2}, [T((0, 0, 1))]_{B_2}]$$

$$\text{and}$$

$$= [[1, 0, 0]^T, (-1, 1, 0)^T, (0, -1, 1)^T]$$

$$\begin{bmatrix} \square \\ \square \\ \square \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

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$$4.3. \text{ RANK-NULLITY THEOREM 75 } 3. \text{ Let } T: \mathbb{R}^3 \rightarrow \mathbb{R}^2 \text{ be defined by } T((x, y, z)) = (x + y - z, x + z). \text{ Let } B_1 = (1, 0, 0), (0, 1, 0), (0, 0, 1) \text{ and } B_2 = (1, 0), (0, 1) \text{ be the ordered bases of the domain and range space,}$$

$$\text{respectively. Then } T[B_1, B_2] = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \text{ Check that that } [T((x, y, z))]_{B_2} = T[B_1, B_2] [(x, y, z)]_{B_1}. \text{ Exercise 4.2.4 Recall the space } P_n(\mathbb{R}) \text{ (the vector space of all polynomials of degree less than or equal to } n \text{). We define a linear transformation } D:$$

$$P_n(\mathbb{R}) \rightarrow P_n(\mathbb{R}) \text{ by}$$

$$D(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = a_1 + 2a_2x + \dots + na_nx^{n-1}. \text{ Find the matrix of the linear transformation } D. \text{ However, note that the image of the linear transformation is contained in } P_{n-1}(\mathbb{R}). \text{ Remark 4.2.5 } 1. \text{ Observe that}$$

$$T[B_1, B_2] = [[T((v_1))]_{B_2}, [T((v_2))]_{B_2}, \dots, [T((v_n))]_{B_2}].$$

$$2. \text{ It is important to note that}$$

$$[T(x)]_{B_2} = T[B_1, B_2] [x]_{B_1}. \text{ That is, we multiply the matrix of the linear transformation with the coordinates } [x]_{B_1} \text{ of the vector } x \in V \text{ to obtain the coordinates of the vector } T(x) \in W. 3. \text{ If } A \text{ is an } m \times n \text{ matrix, then } A \text{ induces}$$

$$\text{a linear transformation } T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m, \text{ defined by}$$

$$76 \text{ CHAPTER 4. LINEAR TRANSFORMATIONS } 3. T \text{ is one-one} \iff N(T) = \{0\} \text{ is the zero subspace of } V \iff \{T(u) : 1 \leq i \leq n\} \text{ is a basis of } 4. \dim(R(T)) = \dim(V) \text{ if and only if } N(T) = \{0\}. \text{ Proof. The results about } R(T)$$

$$\text{and } N(T) \text{ can be easily proved. We thus leave the proof for the readers. We now assume that } T \text{ is one-one. We need to show that } N(T) = \{0\}. \text{ Let } u \in N(T). \text{ Then by definition, } T(u) = 0. \text{ Also for any linear}$$

$$\text{transformation (see Proposition 4.1.3), } T(0) = 0. \text{ Thus } T(u) = T(0). \text{ So, } T \text{ is one-one implies } u = 0. \text{ That is, } N(T) = \{0\}. \text{ Let } N(T) = \{0\}. \text{ We need to show that } T \text{ is one-one. So, let us assume that for some } u, v \in V, T(u) =$$

$$T(v). \text{ Then, by linearity of } T, T(u - v) = 0. \text{ This implies, } u - v \in N(T) = \{0\}. \text{ This in turn implies } u = v. \text{ Hence, } T \text{ is one-one. The other parts can be similarly proved. Remark 4.3.3 } 1. \text{ The space } R(T) \text{ is called the range space}$$

$$\text{of } T \text{ and } N(T) \text{ is called the null space. We write } \rho(T) = \dim(R(T)) \text{ and } \nu(T) = \dim(N(T)). 3. \rho(T) \text{ is called the rank of the linear transformation } T \text{ and } \nu(T) \text{ is called the nullity of } T. \text{ Example 4.3.4 Determine the range and null}$$

$$\text{space of the linear transformation}$$

$$R(T):$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^4 \text{ with } T(x, y, z) = (x - y + z, y - z, x, 2x - 5y + 5z).$$

$$T_A(x) = Ax. \text{ We sometimes write } A \text{ for } T_A. \text{ Suppose that the standard bases for } \mathbb{R}^n \text{ and } \mathbb{R}^m \text{ are the ordered}$$

$$\text{bases } B_1 \text{ and } B_2, \text{ respectively. Then observe that}$$

$$T[B_1, B_2] = A.$$

$$4.3 \text{ Rank-Nullity Theorem Definition 4.3.1 (Range and Null Space) Let } V, W \text{ be finite dimensional vector spaces over}$$

$$\text{the same set of scalars and } T: V \rightarrow W \text{ be a linear transformation. We define } 1. R(T) = \{T(x) : x \in V\}, \text{ and}$$

$$R(T) = L((1, 0, 1, 2), (-1, 1, 0, -5), (1, -1, 0, 5))$$

$$1. (a) R(T) \text{ is a subspace of } W.$$

$$(b) R(T) = L(T(v_1), T(v_2), \dots, T(v_n)).$$

$$\alpha_1$$

$k=1$ a_{1j}

$$\beta^{2..}$$
$$a_{2j..}^{k=1}$$

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- ☐
- ☐
- ☐

$$= I[B_1, B_2] \top [B_1, B_1] I[B_2, B_1] [x]_{B_2}. \quad (4.4.2)$$
$$b_{nk}a_{kj}$$

true for all $x \in V$, we get $[B_1, B_2] T[B_1, B_1] [B_2, B_1] = T[B_2, B_2]$. (4.4.3) That is, $A^{-1}BA = C$ or equivalently $ACA^{-1} = B$. Another

$$b_{ji}u_j \text{ and } T(v_i) = \ast^n$$
 $k=1$
$$C_{kj}V_k = \star^n$$
$$C_{kj}(\star n$$

mark 4.4.9 Observe that if $A = T [B,$

 $k=1$
$$= n$$
 $k=1$

km²

 $0=1$
$$T(v_i) = T(l(v_i)) = T(i)^n$$
$$a_{kj}u_k) = n$$
 $b_k u)$

— n

$$1 + 2x + x^2, 2 + x + x^2. \text{ Then } [1 + x - x^2]_{B_1} = 0 \cdot 1 + 2 \cdot (1 + x) + (-1) \cdot (1 + x + x^2) = (0, 2, -1)^t, [1 + 2x + x^2]_{B_1} = (-1) \cdot 1 + 1 \cdot (1 + x)$$
$$+ 1 \cdot (1 + x + x^2) = (-1, 1, 1)^t, \text{ and}$$

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are said to be similar

 $a_{1k}c_{ki}$

K=

$$a_{2k}c_{kj}$$

Remark 4.4.9 Observe that if $A \equiv T[B, B]$ then

 $c^{2j..}$
$$[T(v_i)]_{B^1} =$$
$$= A$$

1

$$[2 + x + x^2]_{B_1} = 1 \cdot 1 + 0 \cdot (1 + x) + 1 \cdot (1 + x + x^2) = (1, 0, 1)^t.$$
$$I[B_2, B_1] = \left[\prod_{k=1}^n a_{kj} T(u_k) (1+x-x^2) \right]_{B_1}, \left[\prod_{k=1}^n a_{kj} T(u_k) (1+2x+x^2) \right]_{B_1}, \left[\prod_{k=1}^n a_{kj} T(u_k) (2+x+x^2) \right]_{B_1}$$
$$\mathbf{a}_{kj} ({}^n$$
$$= \prod_{k=1}^n [1 + x - x^2]_{B_1}, [1 + 2x + x^2]_{B_1}, [2 + x + x^2]_{B_1} b_k a_k u$$
$$\begin{bmatrix} 0 & -1 & 1 \end{bmatrix}$$

Find the matrices $T[B_1, B_1]$ and $T[B_2, B_2]$. Also verify that $T[B_2 B_2] = I[B_1, B_2] T[B_1, B_1] I[B_2, B_1]$

2. Consider two bases $B_1 = (1, 0, 0), (1, 1, 0), (1, 1, 1)$ and $B_2 = (1, 1, -1), (1, 2, 1), (2, 1, 1)$ of R^3 . Suppose $T: R^3 \rightarrow R^3$ is a linear transformation defined by

$$T((x, y, z)) = (x + y, x + y + 2z, y - z).$$

Find $I[B_1, B_2]$ and verify,

$$I[B_1, B_2] T[B_1, B_1] I[B_2, B_1] = T[B_2, B_2].$$

Check that,

$$T[B_1, B_1] I[B_2, B_1] = I[B_2, B_1] T[B_2, B_2]$$

Exercise 4.4.11 1. Let V be an n -dimensional vector space and let $T: V \rightarrow V$ be a linear transformation. Suppose T has the property that $T^{n-1} = 0$ but $T^n \neq 0$. (a) Then prove that there exists a vector $u \in V$ such that the set $\{u, T(u), \dots, T^{n-1}(u)\}$ is a basis of V . (b) Let $B = (u, T(u), \dots, T^{n-1}(u))$. Then prove that

$$T[B, B] = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

Then

$$I[B, B] T[B, B] I[B, B]^{-1} = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

is the set of all matrices that are similar to the given matrix A . Therefore, similar matrices are just different matrix representations of a single linear transformation.

86 CHAPTER 4. LINEAR TRANSFORMATIONS (c) Let A be an $n \times n$ matrix with the property that $A^{n-1} = 0$ but $A^n \neq 0$. Then prove that A is similar to the matrix given above.

2. Let $T: R^3 \rightarrow R^3$ be a linear transformation given by $T((x, y, z)) = (x + y + 2z, x - y - 3z, 2x + 3y + z)$.

Let B be the standard basis and $B_1 = (1, 1, 1), (-1, 1, 1), (1, 1, 2)$ be another ordered basis. (a) Find the matrices $T[B, B]$ and $T[B_1, B_1]$.

(b) Find the matrix P such that $P^{-1} T[B, B] P = T[B_1, B_1]$.

3. Let $T: R^3 \rightarrow R^3$ be a linear transformation given by $T((x, y, z)) = (x, x + y, x + y + z)$.

Let B be the standard basis and $B_1 = (1, 0, 0), (1, 1, 0), (1, 1, 1)$ be another ordered basis. (a) Find the matrices $T[B, B]$ and $T[B_1, B_1]$.

(b) Find the matrix P such that $P^{-1} T[B, B] P = T[B_1, B_1]$.

4. Let $B_1 = (1, 2, 0), (1, 3, 2), (0, 1, 3)$ and $B_2 = (1, 2, 1), (0, 1, 2), (1, 4, 6)$ be two ordered bases of (a) Find the change of basis matrix P from B_1 to B_2 . (b) Find the change of basis matrix Q from B_2 to B_1 . (c) Verify that $PQ = I = QP$.

(d) Find the change of basis matrix from the standard basis of R^3 to B_1 . What do you notice?

Chapter 5 Inner Product Spaces

We had learned that given vectors $\mathbf{j}_1$ and $\mathbf{j}_2$ (which are at an angle of 90°) in a plane, any vector in the plane is a linear combination of the vectors $\mathbf{j}_1$ and $\mathbf{j}_2$. In this section, we investigate a method by which any basis of a finite dimensional vector can be transferred to another basis in such a way that the vectors

88 CHAPTER 5. INNER PRODUCT SPACES 2. Let $V = C^n$ be a complex vector space of dimension n . Then for $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ in V , check that

$$(u, v) = u_1 v_1 + u_2 v_2 + \dots + u_n v_n = \overline{(v, u)}$$

is an inner product.

3. Let $V = R^2$ and let $A = \begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix}$. Define $(x, y) = xAy^t$. Check that $(\cdot, \cdot)$ is an inner product. Hint: Note that $xAy^t = 4x_1 y_1 - x_1 y_2 - x_2 y_1 + 2x_2 y_2$. 4. Let $x = (x_1, x_2, x_3)$, $y = (y_1, y_2, y_3) \in R^3$. Show that $(x, y) = 10x_1 y_1 + 3x_1 y_2 + 3x_2 y_1 + 2x_2 y_2 + x_3 y_2 + x_3 y_3$ is an inner product in $R^3(R)$.

5. Consider the real vector space R^2 . In this example, we define three products that satisfy two conditions out of the three conditions for an inner product. Hence the three products are not inner products. (a) Define $(x, y) = ((x_1, x_2), (y_1, y_2)) = x_1 y_1$. Then it is easy to verify that the third condition is not valid whereas the first two conditions are valid. (b) Define $(x, y) = ((x_1, x_2), (y_1, y_2)) = x_1^2 + y_1^2 + x_2^2 + y_2^2$. Then it is easy to verify that the first condition is not valid whereas the second and third conditions are valid. (c) Define $(x, y) = ((x_1, x_2), (y_1, y_2)) = x_1 y_1^3 + x_2 y_2^3$. Then it is easy to verify that the second condition is not valid whereas the first and third conditions are valid. Remark 5.1.4 Note that in parts 1 and 2 of Example 5.1.3, the inner products are $u^t v$ and $u v^t$, respectively. This occurs because the vectors u and v are row vectors. In general, u and v are taken as column vectors and hence one uses the notation $u^t v$ or $u \cdot v$. Exercise 5.1.5 Verify that inner products defined in parts 3 and 4 of Example 5.1.3, are indeed inner products. Definition 5.1.6 (Length/Norm of a Vector) For $u \in V$, we define the length (norm) of u , denoted $\|u\|$, in the new basis are at an angle of 90° to each other. To do this, we start by defining a notion of inner product (dot product) in a vector space. This helps us in finding out whether two vectors are at 90° or not.

5.1 Definition and Basic Properties

In R^2 , given two vectors $x = (x_1, x_2)$, $y = (y_1, y_2)$, we know the inner product $x \cdot y = x_1 y_1 + x_2 y_2$. Note that for any $x, y, z \in R^2$ and $\alpha \in R$, this inner product satisfies the conditions

$$x \cdot (y + \alpha z) = x \cdot y + \alpha x \cdot z, \quad x \cdot y = y \cdot x, \quad \text{and} \quad x \cdot x \geq 0 \quad \text{and} \quad x \cdot x = 0 \quad \text{if and only if} \quad x = 0.$$

Thus, we are motivated to define an inner product on an arbitrary

vector space. Definition 5.1.1 (Inner Product) Let $V(F)$ be a vector space over F . An inner product over $V(F)$, denoted by $(\cdot, \cdot)$, is a

map,

such that for $u, v, w \in V$ and $a, b \in F$

- $(u, v) = \overline{(v, u)}$, the complex conjugate of (u, v) , and
- $(u, u) \geq 0$ for all $u \in V$ and equality holds if and only if $u = 0$.

Definition 5.1.2 (Inner Product Space) Let V be a vector space with an

inner product $(\cdot, \cdot)$. Then $(V, (\cdot, \cdot))$ is called an inner product space, in short denoted by ips . Example 5.1.3 The first two examples given

below are called the standard inner product or the dot 1. Let $V = R^n$ be the real vector space of dimension n . Given two vectors $u = (u_1,$

$u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ of V , we define

$$(u, v) = u_1 v_1 + u_2 v_2 + \dots + u_n v_n = u v^t.$$

Verify $(\cdot, \cdot)$ is an inner product.

$$(\cdot, \cdot): V \times V \longrightarrow F$$

5.1. DEFINITION AND BASIC PROPERTIES 89 Or, in other words and the proof of the inequality is over. Observe that if $u = 0$ then the

equality holds if and only of $\lambda u + v = 0$ for $\lambda = -(v, u) \|u\|$

$\|u\|$ (the positive square root. A very useful and a fundamental inequality concerning the inner product is due to Cauchy and Schwartz. The next theorem gives the statement and a proof of this inequality. Theorem 5.1.7 (Cauchy-Schwartz

inequality) Let $V(F)$ be an inner product space. Then for any $u, v \in V$, $|(u, v)| \leq \|u\| \|v\|$. The equality holds if and only if the

vectors u and v are linearly dependent. Further, if $u \neq 0$, then Proof. If $u = 0$, then the inequality holds. Let $u \neq 0$. Note that $(\lambda u$

$$+v, \lambda u + v) \geq 0 \quad \text{for all } \lambda \in F. \quad \text{In particular, for } \lambda = -(v, u) \|u\|^{-2}, \text{ we get}$$

$$0 \leq (\lambda u + v, \lambda u + v)$$

$$1. (au + bv, w) = a(u, w) + b(v, w),$$

90 CHAPTER 5. INNER PRODUCT SPACES4. Let V be a complex vector space with $\dim(V) = n$. Fix an ordered basis $B = (u_1, u_2, \dots, u_n)$. Define a map

$$v = \sum_{i=1}^n \langle v, u_i \rangle u_i$$

and v are linearly dependent. We leave it for the reader to prove product on $\mathbb{R}^n$ and $\mathbb{C}^n$, respectively..

$\langle u, u \rangle^2$. That is, u

$$\langle u, u \rangle^2 \langle u, u \rangle^2 = \langle v, u \rangle^2 \langle v, u \rangle^2$$

$v = \sum_{i=1}^n \langle v, u_i \rangle u_i$ Definition 5.1.8 (Angle between two vectors) Let V be a real vector space. Then for every $u, v \in V$, by the Cauchy-Schwartz inequality, we have

$$87$$

$\langle u, v \rangle \leq 1$. We know that $\cos : [0, \pi] \rightarrow [-1, 1]$ is an one-one and onto function. Therefore, for every real number $\langle u, v \rangle$ and $\langle v, u \rangle$ there exists a unique θ , $0 \leq \theta \leq \pi$, such that

$$\langle u, v \rangle = \langle u, v \rangle \cos \theta$$

1. The real number θ with $0 \leq \theta \leq \pi$ and satisfying $\cos \theta = \langle u, v \rangle / (\|u\| \|v\|)$

$$\langle u, v \rangle \text{ is called the angle between the two vectors } u \text{ and } v.$$

2. The vectors u and v in V are said to be orthogonal if $\langle u, v \rangle = 0$.

whenever $[u]_B = (a_1, a_2, \dots, a_n)^T$ and $[v]_B = (b_1, b_2, \dots, b_n)^T$. Show that the above defined map is indeed an inner product. 5.

Let $x = (x_1, x_2, x_3)$, $y = (y_1, y_2, y_3) \in \mathbb{R}^3$. Show that

$$\langle x, y \rangle = \lambda \langle u, v \rangle + \lambda \langle u, v \rangle + \lambda \langle v, u \rangle + \langle v, v \rangle$$

$$\langle u, v \rangle = \langle v, u \rangle$$

3. A set of vectors $\{u_1, u_2, \dots, u_n\}$ is called mutually orthogonal if $\langle u_i, u_j \rangle = 0$ for all $1 \leq i \neq j \leq n$. Exercise 5.1.9 1. Let $\{e_1, e_2, \dots, e_n\}$ be the standard basis of $\mathbb{R}^n$. Then prove that with respect to the standard inner product on $\mathbb{R}^n$, the vectors e_i satisfy the following:

(a) $\langle e_i, e_i \rangle = 1$ for $1 \leq i \leq n$.

(b) $\langle e_i, e_j \rangle = 0$ for $1 \leq i \neq j \leq n$.

2. Recall the following inner product on $\mathbb{R}^2$: for $x = (x_1, x_2)^T$ and $y = (y_1, y_2)^T$,

$$\langle x, y \rangle = 4x_1y_1 - x_1y_2 - x_2y_1 + 2x_2y_2.$$

(a) Find the angle between the vectors $e_1 = (1, 0)^T$ and $e_2 = (0, 1)^T$.

(b) Let $u = (1, 0)^T$. Find $v \in \mathbb{R}^2$ such that $\langle v, u \rangle = 0$.

(c) Find two vectors $x, y \in \mathbb{R}^2$, such that $\langle x, x \rangle = \langle y, y \rangle = 1$ and $\langle x, y \rangle = 0$.

3. Find an inner product in $\mathbb{R}^2$ such that the following conditions hold: $\langle (1, 2), (2, -1) \rangle = 1$, and $\langle (1, 2), (2, -1) \rangle = 0$. [Hint: Consider a symmetric matrix $A = a$ b

vectors u and v in V .

b c . Define $\langle x, y \rangle = y^T A x$ and solve a system of 3 equations for the unknowns a, b, c .]

8. Let V be an inner product space. Prove that

$\langle u + v, u + v \rangle \leq \langle u, u \rangle + \langle v, v \rangle$ for every $u, v \in V$. This inequality is called the triangle inequality.

$(1, 1, 1)$ and $(2, -5, 2)$.

9. Let $z_1, z_2, \dots, z_n \in \mathbb{C}$. Use the Cauchy-Schwartz inequality to prove that $|z_1 + z_2 + \dots + z_n| \leq \sqrt{n(|z_1|^2 + |z_2|^2 + \dots + |z_n|^2)}$. When does the equality hold? 10. Let $x, y \in \mathbb{R}^n$. Observe that $\langle x, y \rangle = \langle y, x \rangle$. Hence or otherwise prove the following: (a) $\langle x, y \rangle = 0 \iff \langle x - y, x - y \rangle = \langle x, x \rangle + \langle y, y \rangle$. (This is called Pythagoras Theorem). (b) $\langle x, x \rangle = \langle y, y \rangle \iff \langle x + y, x - y \rangle = 0$, (x and y form adjacent sides of a rhombus as the diagonals

$$\langle A A^T \rangle_{ii} = \sum_{j=1}^n a_{ij}^2$$

$$\langle A, A \rangle = \sum_{i=1}^n \langle A_i, A_i \rangle = \sum_{i=1}^n \|A_i\|^2$$

$x + y$ and $x - y$ are orthogonal).

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 = \sum_{i=1}^n \sum_{j=1}^n a_{ji}^2$$

(c) $\langle x + y, x + y \rangle + \langle x - y, x - y \rangle = 2\langle x, x \rangle + 2\langle y, y \rangle$. (This is called the Parallelogram Law). (d) $4\langle x, y \rangle = \langle x + y, x + y \rangle - \langle x - y, x - y \rangle$ (This is called the polarisation identity). Remark 5.1.10 i. Suppose the norm of a vector is given. Then,

the polarisation identity can be used to define an inner product. $M_{n \times n}(\mathbb{R})$.

5.1. DEFINITION AND BASIC PROPERTIES 91 ii. Observe that if $\langle x, y \rangle = 0$ then the parallelogram spanned by the vectors x and y is a rectangle. The above equality tells us that the lengths of the two diagonals are

equal. Are these results true if $x, y \in \mathbb{C}^n(C)$? 11. Let $x, y \in \mathbb{C}^n(C)$. Prove that

(b) If $x \neq 0$ then $\langle x + ix, x + ix \rangle = \langle x, x \rangle + \langle ix, ix \rangle$, even though $\langle x, ix \rangle \neq 0$.

(c) If $\langle x + y, x + y \rangle = \langle x, x \rangle + \langle y, y \rangle$ and $\langle x + iy, x + iy \rangle = \langle x, x \rangle + \langle y, y \rangle$ then show that $\langle x, y \rangle = 0$. 12. Let V be an n -dimensional inner product space, with an inner product $\langle \cdot, \cdot \rangle$. Let $u \in V$ be a fixed vector with $\langle u, u \rangle = 1$. Then give reasons for the following statements. (a) Let

$$S^\perp = \{v \in V : \langle v, u \rangle = 0\}.$$

Then S is a subspace of V of dimension $n - 1$. (b) Let $0 \neq \alpha \in F$ and let $S = \{v \in V : \langle v, u \rangle = \alpha\}$. Then S is not a

subspace of V .

(c) For any $v \in S$, there exists a vector $v_0 \in S^\perp$, such that $v = v_0 + \alpha u$. Theorem 5.1.11 Let V be an inner product space. Let $\{u_1, u_2, \dots, u_n\}$ be a set of non-zero, mutually orthogonal vectors of V . 1. Then the set $\{u_1, u_2, \dots, u_n\}$ is linearly independent.

3. Let $\dim(V) = n$ and also let $\langle u_i, u_i \rangle = 1$ for $i = 1, 2, \dots, n$. Then for any $v \in V$, In particular, $\langle v, u_i \rangle = 0$ for all $i = 1, 2, \dots, n$ if and only if $v = 0$.

(a) $4\langle x, y \rangle = \langle x + y, x + y \rangle - \langle x - y, x - y \rangle + \langle x + iy, x + iy \rangle - \langle x - iy, x - iy \rangle$. Proof. Consider the set of non-zero, mutually orthogonal vectors $\{u_1, u_2, \dots, u_n\}$. Suppose there exist

scalars $c_1, c_2, \dots, c_n$ not all zero, such that

$c_1 u_1 + c_2 u_2 + \dots + c_n u_n = 0$. Then for $1 \leq i \leq n$, we have $0 = \langle 0, u_i \rangle = \langle c_1 u_1 + c_2 u_2 + \dots + c_n u_n, u_i \rangle = \sum_{j=1}^n c_j \langle u_j, u_i \rangle = c_i$ as $\langle u_i, u_i \rangle = 1$ and $\langle u_i, u_j \rangle = 0$ for all $j \neq i$ and $\langle u_i, u_i \rangle = 1$. This gives a contradiction to our assumption that some of the c_i 's are

non-zero. This establishes the linear independence of a set of non-zero, mutually orthogonal $\|u_i\|$

Figure 5.1: Gram-Schmidt Process

Theorem 5.2.1 (Gram-Schmidt Orthogonalization Process)

Let V be an inner product space. Suppose

vectors. For the second part, using $\langle u_i, u_j \rangle = 0$ if $i \neq j$ and $\langle u_i, u_i \rangle = 1$ for $1 \leq i, j \leq n$, we have

2. $\|u_i\|^2 = 1$

$$\langle u_i, u_i \rangle^2 = 1^n$$

$$\|u_i\|^2 \langle u_i, u_i \rangle^2 = 1^n$$

$i=1$

$$v = \sum_{i=1}^n (v, u_i) u_i$$

5.2. GRAM-SCHMIDT ORTHOGONALISATION PROCESS 93

$$c_j(u_i, u_j) = c_{ij}$$

$$\begin{aligned} & \sum_{i=1}^n \alpha_i u_i \\ & \left(\sum_{i=1}^n \alpha_i u_i \right)^2 = \sum_{i=1}^n \alpha_i^2 u_i^2 \\ & \sum_{i=1}^n \alpha_i (u_i, \sum_{j=1}^n \alpha_j u_j) \\ & = \sum_{i=1}^n \alpha_i^2 \\ & \sum_{i=1}^n \alpha_i (u_i, u_j) = \sum_{i=1}^n \alpha_i \alpha_j (u_i, u_j) \\ & = \sum_{i=1}^n \alpha_i^2 \end{aligned}$$

$\{u_1, u_2, \dots, u_n\}$ is a set of linearly independent vectors of V . Then there exists a set $\{v_1, v_2, \dots, v_n\}$ of vectors of V satisfying the following: 1. $(v_i, v_j) = \delta_{ij}$ for $1 \leq i, j \leq n$.

2. $(v_i, v_j) = 0$ for $1 \leq i < j \leq n$.

3. $L(v_1, v_2, \dots, v_i) = L(u_1, u_2, \dots, u_i)$ for $1 \leq i \leq n$.

Proof. We successively define the vectors $v_1, v_2, \dots, v_n$ as follows:

$$v_1 = \frac{1}{\|u_1\|} u_1$$

$$v_i = \frac{1}{\|u_i\|} u_i$$

Calculate $w_2 = u_2 - (u_2, v_1)v_1$, and let $v_2 = w_2 / \|w_2\|$.

92 CHAPTER 5. INNER PRODUCT SPACES For the third part, observe from the first part, the linear independence of the non-zero mutually orthogonal vectors $u_1, u_2, \dots, u_n$. Since $\dim(V) = n$, they form a basis of V .

Thus, for every vector $v \in V$, there exist scalars α_i , $1 \leq i \leq n$, such that $v = \sum_{i=1}^n \alpha_i u_i$. Hence, $\alpha_i (u_i, u_j) = \alpha_j \delta_{ij}$. Therefore, we have obtained the required result. Definition 5.1.12 (Orthonormal Set) Let V be an inner product space. A set of non-zero, mutually orthogonal vectors $\{v_1, v_2, \dots, v_n\}$ in V is called an orthonormal set if $(v_i, v_j) = \delta_{ij}$ for $i, j = 1, 2, \dots, n$. If the set $\{v_1, v_2, \dots, v_n\}$ is also a basis of V , then the set of vectors $\{v_1, v_2, \dots, v_n\}$ is called an orthonormal basis of V .

Example 5.1.13 1. Consider the vector space $\mathbb{R}^2$ with the standard inner product. Then the standard ordered basis $B = \{(1, 0), (0, 1)\}$ is an orthonormal set. Also, the basis $B_1 = \left\{ \frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(1, -1) \right\}$ is an orthonormal set. 2. Let $\mathbb{R}^n$ be endowed with the standard inner product. Then by Exercise 5.1.9.1, the standard ordered basis $B = \{e_1, e_2, \dots, e_n\}$ is an orthonormal set. In view of Theorem 5.1.11, we inquire into the question of extracting an orthonormal basis from a given basis. In the next section, we describe a process (called the Gram-Schmidt Orthogonalisation process) that generates an orthonormal set from a given set containing finitely many vectors.

Remark 5.1.14 The last part of the above theorem can be rephrased as "suppose $\{v_1, v_2, \dots, v_n\}$ is an orthonormal basis of an inner product space V . Then for each $u \in V$ the numbers (u, v_i) for $1 \leq i \leq n$ are the coordinates of u with respect to the above basis". That is, let $B = (v_1, v_2, \dots, v_n)$ be an ordered basis. Then for any $u \in V$,

$$(v, u) = \sum_{i=1}^n \alpha_i (v, u_i) = \sum_{i=1}^n \alpha_i^2$$

$$(u)_B = ((u, v_1), (u, v_2), \dots, (u, v_n))$$

5.2 Gram-Schmidt Orthogonalisation Process Let V be a finite dimensional inner product space. Suppose $u_1, u_2, \dots, u_n$ is a linearly independent subset of V . Then the Gram-Schmidt orthogonalisation process uses the vectors $u_1, u_2, \dots, u_n$ to construct new vectors $v_1, v_2, \dots, v_n$ such that $(v_i, v_j) = \delta_{ij}$ for $i, j = 1, 2, \dots, n$ and $\text{Span}\{u_1, u_2, \dots, u_i\} = \text{Span}\{v_1, v_2, \dots, v_i\}$ for $i = 1, 2, \dots, n$. This process proceeds with the following idea. Suppose we are given two vectors u and v in a plane. If we want to get vectors z and y such that z is a unit vector in the direction of u and y is a unit vector perpendicular to z , then they can be obtained in the following way:

Take the first vector $z = \frac{u}{\|u\|}$. Let θ be the angle between the vectors u and v . Then $\cos(\theta) = \frac{(u, v)}{\|u\| \|v\|}$. Defined $\alpha = \|v\| \cos(\theta) = (u, v)$.

$w = (z, v)$. Then $w = v - \alpha z$ is a vector perpendicular to the unit vector z , as we have removed the component of v from z . So, the vectors that we are interested in are w and z . This idea is used to give the Gram-Schmidt Orthogonalisation process which we now describe.

94 CHAPTER 5. INNER PRODUCT SPACES 3. Let $(v_1, v_2, \dots, v_i) = L(u_1, u_2, \dots, u_i)$ for $1 \leq i \leq k$. Now, let us assume that we are given a set of n linearly independent vectors $\{u_1, u_2, \dots, u_n\}$ of V . Then by the inductive assumption, we already have vectors $v_1, v_2, \dots, v_{n-1}$ satisfying 1. $(v_i, v_j) = \delta_{ij}$ for $1 \leq i, j \leq n-1$,

2. $(v_i, v_j) = 0$ for $1 \leq i < j \leq n-1$, and

3. $L(v_1, v_2, \dots, v_i) = L(u_1, u_2, \dots, u_i)$ for $1 \leq i \leq n-1$. Using (5.2.1), we define

z and $y = w$

$w_n = u_n - (u_n, v_1)v_1 - (u_n, v_2)v_2 - \dots - (u_n, v_{n-1})v_{n-1}$. (5.2.2) We first show that $w_n \neq 0$. This will also imply that $w_n \neq 0$ and hence $v_n = \frac{w_n}{\|w_n\|}$ is well defined. On the contrary, assume that $w_n \in L(v_1, v_2, \dots, v_{n-1})$. Then there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_{n-1}$ such that $w_n = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{n-1} v_{n-1}$. So, by (5.2.2) $u_n = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{n-1} v_{n-1} + (u_n, v_1)v_1 + (u_n, v_2)v_2 + \dots + (u_n, v_{n-1})v_{n-1}$. Thus, by the third induction assumption, $u_n \in L(v_1, v_2, \dots, v_{n-1}) = L(u_1, u_2, \dots, u_{n-1})$. This gives a contradiction to the given assumption that the set of vectors $\{u_1, u_2, \dots, u_n\}$ is linearly independent.

Obtain $w_3 = u_3 - (u_3, v_1)v_1 - (u_3, v_2)v_2$, and let $v_3 = \frac{w_3}{\|w_3\|}$. In general, if $v_1, v_2, v_3, \dots, v_{i-1}$ are already obtained, we compute $w_i = u_i - (u_i, v_1)v_1 - (u_i, v_2)v_2 - \dots - (u_i, v_{i-1})v_{i-1}$. (5.2.1) and define $v_i = \frac{w_i}{\|w_i\|}$. We prove the theorem by induction on n , the number of linearly independent vectors. For $n = 1$, we have $v_1 = u_1$.

$w_n = 0$. Then $v_n = 1$. Also, it can be easily verified that $(v_i, v_j) = 0$ for $1 \leq i \leq n-1$. Hence, by the principle of mathematical induction, the proof of the theorem is complete. We illustrate the Gram-Schmidt process by the following example. **Example 5.2.2** Let $\{(1, -1, 1, 1), (1, 0, 1, 0), (0, 1, 0, 1)\}$ be a linearly independent set in $\mathbb{R}^4$. Find an orthonormal set $\{v_1, v_2, v_3\}$ such that $L(\{(1, -1, 1, 1), (1, 0, 1, 0), (0, 1, 0, 1)\}) = L(v_1, v_2, v_3)$.

So, $w_n = 0$. Define $v_n = w_n$.

Solution: Let $u_1 = (1, 0, 1, 0)$. Define $v_1 = (1, 0, 1, 0)$.

$\sqrt{2}$. Let $u_2 = (0, 1, 0, 1)$. Then

$$w_2 = (0, 1, 0, 1) - ((0, 1, 0, 1), (1, 0, 1, 0)) \cdot (1, 0, 1, 0)$$

$\sqrt{2} = 1$. Hence, the result holds for $n = 1$. Let the result hold for all $k \leq n-1$. That is, suppose we are given any set of k , $1 \leq k \leq n-1$ linearly independent vectors $\{u_1, u_2, \dots, u_k\}$ of V . Then by the inductive

assumption, there exists a set $\{v_1, v_2, \dots, v_k\}$ of vectors satisfying the following:

$$v_1 = u_1$$

$\&u_1$.

1. $(v_i, v_j) = 1$ for $1 \leq i \leq k$,

2. $(v_i, v_j) = 0$ for $1 \leq i \leq j \leq k$, and

$$\sqrt{2} v_1 = (0, 1, 0, 1).$$

$\sqrt{2}$. Let $u_3 = (1, -1, 1, 1)$. Then

$$w_3 = (1, -1, 1, 1) - ((1, -1, 1, 1), (1, 0, 1, 0)) \cdot (1, 0, 1, 0) - ((1, -1, 1, 1), (0, 1, 0, 1)) \cdot (0, 1, 0, 1) = (0, -1, 0, 1)$$

$$\&w_3.$$

Hence,

$$v_2 = (0, 1, 0, 1)$$

$$\&u_1. \text{ Since } u_1 = 0, v_1 = 0 \text{ and } \&u_1 = (u_1, u_1)$$

and v_3

$$= (0, -1, 0, 1) \cdot \sqrt{2}.$$

$$\&v_1^2 = (v_1, v_1) = (u_1, u_1) = \&u_1.$$

independent.

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Remark 5.2.3 1. Let $\{u_1, u_2, \dots, u_k\}$ be any basis of a k -dimensional subspace W of $\mathbb{R}^n$. Then by Gram-Schmidt orthogonalisation process, we get an

orthonormal set $\{v_1, v_2, \dots, v_k\} \subset \mathbb{R}^n$ with

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Note that,

$$v_1^2 = (v_1, v_1) = (u_1, u_1)$$

$$W = L(v_1, v_2, \dots, v_k), \text{ and for } 1 \leq i \leq k,$$

$$L(v_1, v_2, \dots, v_i) = L(u_1, u_2, \dots, u_i).$$

2. Suppose we are given a set of n vectors, $\{u_1, u_2, \dots, u_n\}$ of V that are linearly dependent. Then by Corollary 3.2.5, there exists a smallest k , $2 \leq k \leq n$ such that $L(u_1, u_2, \dots, u_k) = L(u_1, u_2, \dots, u_{k-1})$. We claim that in

this case, $w_k = 0$. Since, we have chosen the smallest k satisfying

$$L(u_1, u_2, \dots, u_i) = L(u_1, u_2, \dots, u_{i-1}),$$

for $2 \leq i \leq n$, the set $\{u_1, u_2, \dots, u_{k-1}\}$ is linearly independent (use Corollary 3.2.5). So, by Theorem 5.2.1, there exists an orthonormal set $\{v_1, v_2, \dots, v_{k-1}\}$ such that $L(u_1, u_2, \dots, u_{k-1}) = L(v_1, v_2, \dots, v_{k-1})$.

As $u_k \in L(v_1, v_2, \dots, v_{k-1})$, by Remark 5.1.14

$$u_k = (u_k, v_1)v_1 + (u_k, v_2)v_2 + \dots + (u_k, v_{k-1})v_{k-1}. \text{ So, by definition of } w_k, w_k = 0. \text{ Therefore, in this case, we can continue with the Gram-Schmidt process by replacing } u_k \text{ by } u_{k+1}.$$

$$\{v_1, v_2, \dots, v_k\}$$

$$v_2, v_1, v_2^2, \dots, v_2, v_k$$

$$v_k, v_1, v_k, v_2, \dots, v_k$$

$$1 \ 0 \ \dots \ 0$$

$$0 \ 1 \ \dots \ 0$$

$$0 \ 0 \ \dots \ 1 \ \dots \ 0 = I_k. \text{ Or using (5.2.3), in the language of matrices, we get } \alpha_{1k} \alpha_{2k} \dots \alpha_{rk} \alpha_{11} \alpha_{12} \dots \alpha_{1k} \alpha_{21} \alpha_{22} \dots \alpha_{2k} \dots \alpha_{rk} \alpha_{r1} \alpha_{r2} \dots \alpha_{rk} = I_k.$$

$$\alpha_{11} \alpha_{21} \dots \alpha_{n1}$$

$$\alpha_{12} \alpha_{22} \dots \alpha_{n2}$$

$$\alpha_{1n} \alpha_{2n} \dots \alpha_{nn} = I_n.$$

Perhaps the readers must have noticed that the inverse of A is its transpose. Such matrices are called orthogonal matrices and they have a special role to play. **Definition 5.2.4** (Orthogonal Matrix)

An $n \times n$ real matrix A is said to be an orthogonal matrix if $A^t = A^{-1}$. It is worthwhile to solve the following exercises. **Exercise 5.2.5** 1. Let A and B be two $n \times n$ orthogonal matrices. Then prove that AB and BA are both

orthogonal matrices. 2. Let A be an $n \times n$ orthogonal matrix. Then prove that (a) the rows of A form an orthonormal basis of $\mathbb{R}^n$.

$$\square$$

$$\square$$

$$\square$$

$$v_1$$

$$v_2$$

$$\square$$

$$\square$$

$$\square$$

$$A^t A =$$

$$=$$

$$\square$$

$$\square$$

$$\square$$

$$\square$$

$$\square$$

3. Let S be a countably infinite set of linearly independent vectors. Then one can apply the Gram-Schmidt process to get a countably infinite orthonormal set.

4. Let $\{v_1, v_2, \dots, v_k\}$ be an orthonormal subset of $\mathbb{R}^n$. Let $B = (e_1, e_2, \dots, e_n)$ be the standard ordered basis of

$\mathbb{R}^n$. Then there exist real numbers α_{ij} , $1 \leq i \leq k$, $1 \leq j \leq n$ such that $[v_i]_B = (\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{in})^t$.

Let $A = [v_1, v_2, \dots, v_n]$. Then in the ordered basis B , we have

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

is an $n \times k$ matrix. Also, observe that the conditions $\|v_i\| = 1$ and $\langle v_i, v_j \rangle = 0$ for $1 \leq i \neq j \leq n$, implies that $1 = \|v_i\|^2 = \langle v_i, v_i \rangle = \langle v_i, \sum_{j=1}^n v_j \rangle = \langle v_i, v_i \rangle + \sum_{j \neq i} \langle v_i, v_j \rangle = 1 + 0 = 1$.

$$\langle v_i, v_j \rangle = \delta_{ij}$$

$$\text{and } 0 = \langle v_i, v_j \rangle = \delta_{ij}$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

(b) the columns of A form an orthonormal basis of $\mathbb{R}^n$.

(c) for any two vectors $x, y \in \mathbb{R}^{n \times 1}$, $\langle Ax, Ay \rangle = \langle x, y \rangle$. (d) for any vector $x \in \mathbb{R}^{n \times 1}$, $\|Ax\| = \|x\|$.

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Theorem 5.2.6 (QR Decomposition) Let A be a square matrix of order n . Then there exist matrices Q

and R such that Q is orthogonal and R is upper triangular with $A = QR$.

In case, A is non-singular, the diagonal entries of R can be chosen to be positive. Also, in this case, the decomposition is unique.

$$A =$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

Proof. We prove the theorem when A is non-singular. The proof for the singular case is left as an exercise. Let the columns of A be $x_1, x_2, \dots, x_n$. The Gram-Schmidt orthogonalisation process applied to the vectors $x_1, x_2, \dots, x_n$ gives the vectors $u_1, u_2, \dots, u_n$ satisfying

$$L(u_1, u_2, \dots, u_n) = L(x_1, x_2, \dots, x_n),$$

$$\|u_i\| = 1, \langle u_i, u_j \rangle = 0, \text{ for } 1 \leq i \neq j \leq n. \quad (5.2.4)$$

Now, consider the ordered basis $B = (u_1, u_2, \dots, u_n)$. From (5.2.4), for $1 \leq i \leq n$, we have $L(u_1, u_2, \dots, u_i) =$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1i} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2i} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{i1} & \alpha_{i2} & \cdots & \alpha_{ii} \end{pmatrix} \quad (5.2.3)$$

$L(x_1, x_2, \dots, x_i)$. So, we can find scalars α_{ij} , $1 \leq j \leq i$ such that $x_i = \alpha_{i1}u_1 + \alpha_{i2}u_2 + \cdots + \alpha_{ii}u_i = \begin{pmatrix} \alpha_{i1} & \alpha_{i2} & \cdots & \alpha_{ii} & 0 & \cdots & 0 \end{pmatrix}_B$. (5.2.5)

3. R is an $n \times k$ matrix with $\text{rank}(R) = r$.

Example 5.2.8 1. Let $A = \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 1 & -1 & 1 \end{pmatrix}$

Let $Q = [u_1, u_2, \dots, u_n]$. Then by Exercise 5.2.5.3, Q is an orthogonal matrix. We now define an $n \times n$

upper triangular matrix R by

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ 0 & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{nn} \end{pmatrix}$$

By using (5.2.5), we get

$$QR = [u_1, u_2, \dots, u_n] \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ 0 & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{nn} \end{pmatrix}$$

matrix R such that $A = QR$. Solution: From Example 5.2.2, we

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ 0 & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{nn} \end{pmatrix}$$

$$\alpha_{11}u_1, \alpha_{12}u_1 + \alpha_{22}u_2, \dots, \alpha_{nn}u_n$$

$[x_1, x_2, \dots, x_n] = A$. Thus, we see that $A = QR$, where Q is an orthogonal matrix (see Remark 5.2.3.4) and R is an upper triangular matrix. The proof doesn't guarantee that for $1 \leq i \leq n$, α_{ii} is positive. But this can be

achieved by replacing the vector u_i by $-u_i$ whenever α_{ii} is negative.

1. Observe the following properties of upper triangular matrices.

where 1. The inverse of an upper triangular matrix is also an upper triangular matrix, and

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ 0 & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{nn} \end{pmatrix}^{-1} = \begin{pmatrix} \alpha_{11}^{-1} & -\alpha_{11}^{-1}\alpha_{12}\alpha_{22}^{-1} & \cdots & -\alpha_{11}^{-1}\alpha_{1n}\alpha_{22}^{-1}\alpha_{33}^{-1}\cdots\alpha_{nn}^{-1} \\ 0 & \alpha_{22}^{-1} & \cdots & -\alpha_{22}^{-1}\alpha_{2n}\alpha_{33}^{-1}\cdots\alpha_{nn}^{-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{nn}^{-1} \end{pmatrix}$$

2. product of upper triangular matrices is also upper triangular. Thus the matrix $R_2R_1^{-1}$

$$R =$$

is an upper triangular matrix. Also, by Exercise 5.2.5.1, the matrix Q^{-1}

$$Q^{-1}$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

is an orthogonal matrix. Hence, by Exercise 5.2.5.4, $R_2R_1^{-1}$

The readers are advised to check that $A = QR$ is indeed correct. 2. Let $A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{pmatrix}$

Find a 4×3 matrix Q satisfying $Q^TQ = I_3$ and an upper triangular matrix

R_1 . So, $R_2 = R_1$ and therefore $Q_2 = Q_1$. Suppose we have matrix $A = [x_1, x_2, \dots, x_k]$ of dimension $n \times k$ with $\text{rank}(A) = r$. Then by Remark

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

5.2.3.2, the application of the Gram-Schmidt orthogonalisation process yields a set $\{u_1, u_2, \dots, u_r\}$ of

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

$$6$$

$$=$$

$$\sum_{j=1}^r \alpha_{ij}u_j$$

Uniqueness: suppose $Q_1R_1 = Q_2R_2$ then Q^{-1}

$$Q_1R_1 = Q_2R_2$$

R such that $A = QR$.

Solution: Let us apply the Gram Schmidt orthogonalisation to the columns of A . Or equivalently to the

$$w_4 = u_4 - \langle u_4, v_1 \rangle v_1 - \langle u_4, v_2 \rangle v_2 - \langle u_4, v_3 \rangle v_3$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

3. Let $\{u_1, u_2, \dots, u_n\}$ be an orthonormal basis of $\mathbb{R}^n$. Let $B = (e_1, e_2, \dots, e_n)$ be the standard basis of

$\mathbb{R}^n$. Construct an $n \times n$ matrix A by

$$A^T A =$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

$$A = [u_1, u_2, \dots, u_n] = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} & \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} & \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nn} \end{pmatrix}$$

$$A^T A = I_n.$$

$$a_{ij}e_j, \text{ for } 1 \leq i \leq n.$$

Prove that $A^T A = I_n$. Hence deduce that A is an orthogonal matrix.

4. Let A be an $n \times n$ upper triangular matrix. If

A is also an orthogonal matrix, then prove that $A = I_n$.

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orthonormal vectors of $\mathbb{R}^n$. In this case, for each i , $1 \leq i \leq r$, we have $L(u_1, u_2, \dots, u_i) = L(x_1, x_2, \dots, x_i)$, for some j , $i \leq j \leq k$. Hence, proceeding on the lines of the above theorem, we have the following

Theorem 5.2.7 (Generalised QR Decomposition) Let A be an $n \times k$ matrix of rank r . Then $A = QR$,

where

1. Q is an $n \times r$ matrix with $Q^T Q = I_r$. That is, the columns of Q form an orthonormal set.

2. If $Q = [u_1, u_2, \dots, u_r]$, then

$$L(u_1, u_2, \dots, u_i) = L(x_1, x_2, \dots, x_i), \text{ and } j = i$$

Find an orthogonal matrix Q and an upper triangular

$$Q = \frac{1}{2} \begin{pmatrix} \sqrt{2} & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

of $\mathbb{R}^4$.

5.2. GRAM-SCHMIDT ORTHOGONALISATION PROCESS 99 Let $u_1 = (1, -1, 1, 1)$. Define $v_1 = \frac{1}{2}u_1$. Let $u_2 = (1, 0, 1, 0)$. Then

$$w_2 = (1, 0, 1, 0) - (u_2, v_1)v_1 = (1, 0, 1, 0) - v_1 = \frac{1}{2}(1, 1, 1, -1).$$

Hence, $v_2 = \frac{1}{2}(1, 1, 1, -1)$. Let $u_3 = (1, -2, 1, 2)$. Then

$$w_3 = u_3 - (u_3, v_1)v_1 - (u_3, v_2)v_2 = u_3 - 3v_1 + v_2 = 0. \text{ So, we again take } u_3 = (0, 1, 0, 1). \text{ Then } \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 2 & 0 \end{pmatrix}$$

$$Q = [v_1, v_2, v_3] = \frac{1}{2} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The readers are advised to check the following: (a) $\text{rank}(A) = 3$, (b) $A = QR$ with $Q^T Q = I_3$, and

(c) R a 3×4 upper triangular matrix with $\text{rank}(R) = 3$.

Exercise 5.2.9 1. Determine an orthonormal basis of $\mathbb{R}^4$ containing the vectors $(1, -2, 1, 3)$ and $(2, 1, -3, 1)$. 2. Prove that the polynomials $1, x, x^2, x^3$ form an orthogonal set of functions in the inner product space $C[-1, 1]$ with the inner product $(f, g) = \int_{-1}^1 f(t)g(t)dt$. Find the corresponding functions, $f(x)$ with $\int_{-1}^1 f(x)dx = 1$.

3. Consider the vector space $C[-\pi, \pi]$ with the standard inner product defined in the above exercise. Find an orthonormal basis for the subspace spanned by $x, \sin x$ and $\sin(x + 1)$.

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space V of dimension n , one can always

find an $(n - k)$ -dimensional vector subspace W_0 of V (see Exercise 3.3.19.9) satisfying $W + W_0 = V$ and $W \cap W_0 = \{0\}$. The subspace W_0 is called the

complementary subspace of W in V . We now define an important class of linear transformations on an inner product space, called orthogonal projections. Definition 5.3.1 (Projection Operator) Let V be an n -dimensional

vector space and let W be a k -dimensional subspace of V . Let W_0 be a complement of W in V . Then we define a map $P_W: V \rightarrow V$ by $P_W(v) = w$, whenever $v = w + w_0$, $w \in W$, $w_0 \in W_0$. The map P_W is

called the projection of V onto W along W_0 . Remark 5.3.2 The map P is well defined due to the following reasons: 1. $W + W_0 = V$ implies that for every $v \in V$, we can find $w \in W$ and $w_0 \in W_0$ such that $v = w + w_0$. 2. $W \cap W_0 = \{0\}$ implies that the expression $v = w + w_0$ is unique for every $v \in V$. The next proposition states that the map defined above is a linear transformation from V to V . We omit the proof, as it follows directly from the above

remarks. Proposition 5.3.3 The map $P_W: V \rightarrow V$ defined above is a linear transformation. Example 5.3.4 Let $V = \mathbb{R}^3$ and $W = \{(x, y, z) \in \mathbb{R}^3: x + y - z = 0\}$. 1. Let $W_0 = L(\{(1, 2, 2)\})$. Then $W \cap W_0 = \{0\}$ and $W + W_0 = \mathbb{R}^3$.

Also, for any vector $(x, y, z) \in \mathbb{R}^3$ note that $(x, y, z) = w + w_0$, where $w = (z - y, 2z - 2x - y, 3z - 2x - 2y)$, and $w_0 = (x + y - z)(1, 2, 2)$. So, by definition,

4. Let M be a subspace of $\mathbb{R}^3$ and $\dim M = m$. A vector $x \in \mathbb{R}^3$ is said to be orthogonal to M if $w_3 = u_3 - (u_3, v_1)v_1 - (u_3, v_2)v_2 = u_3 - 0v_1 - 0v_2 = u_3$.

$(x, y) = 0$ for every $y \in M$.

So, $v_3 = (0, 1, 0, 1)$

(a) How many linearly independent vectors can be orthogonal to M ?

2. Hence,

(b) If $M = \{(x_1, x_2, x_3) \in \mathbb{R}^3: x_1 + x_2 + x_3 = 0\}$, determine a maximal set of linearly independent vectors orthogonal to M in $\mathbb{R}^3$.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

5. Determine an orthogonal basis of vector subspace spanned by

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 2 & -1 & 2 \end{pmatrix}$$

6. Let $S = \{(1, 1, 1, 1), (1, 2, 0, 1), (2, 2, 4, 0)\}$. Find an orthonormal basis of $L(S)$ in $\mathbb{R}^4$.

$$P_W((x, y, z)) = (z - y, 2z - 2x - y, 3z - 2x - 2y) = \begin{pmatrix} 0 & 0 & 1 \\ -1 & -2 & -1 \\ -2 & -2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

2. Let $W_0 = L(\{(1, 1, 1)\})$. Then $W \cap W_0 = \{0\}$ and $W + W_0 = \mathbb{R}^3$. Also, for any vector $(x, y, z) \in \mathbb{R}^3$ note that $(x, y, z) = w + w_0$,

W_0 is unique if we put an additional condition that $W_0 = \{v \in V: (v, w) = 0 \text{ for all } w \in W\}$. We now prove some basic properties about projection maps. Theorem 5.3.6 Let W and W_0 be complementary subspaces of a vector space V . Let $P_W: V \rightarrow V$ be a projection operator of V onto W along W_0 . Then

1. the null space of P_W , $N(P_W) = \{v \in V: P_W(v) = 0\} = W_0$. 2. the range space of P_W , $R(P_W) = \{P_W(v): v \in V\} = W$.

Proof. We leave the prove of the first part for the reader. The prove of the second part is as follows: Let $\dim(V) = n$ and $\dim(W) = k$.

3. $P_W^2 = P_W$. The condition $P_W^2 = P_W$ is equivalent to $P_W(I - P_W) = 0 = (I - P_W)P_W$.

Proof. We only prove the first part of the theorem.

subspace W_0 .

Let $w_0 \in W_0$. Then $w_0 = 0 + w_0$ for $0 \in W$. So, by definition, $P(w_0) = 0$. Hence, $W_0 \subset N(P_W)$. Also, for any $v \in V$, let $P_W(v) = 0$ with $v = w + w_0$ for some $w_0 \in W_0$ and $w \in W$. Then by

definition $0 = P_W(v) = w$. That is, $w = 0$ and $v = w_0$. Thus, $v \in W_0$. Hence $N(P_W) = W_0$. Exercise 5.3.7 1. Let A be an $n \times n$ real matrix with $A^2 = A$. Consider the linear transformation

Recall that given a k -dimensional vector subspace of a vector

7. Let $\mathbb{R}^n$ be endowed with the standard inner product. Suppose we have a vector $x^T = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ with $\|x\| = 1$. Then prove

the following: (a) the set $\{x\}$ can always be extended to form an orthonormal basis of $\mathbb{R}^n$.

$$P_W((x, y, z)) = (z - y, 2z - 2x - y) = \begin{pmatrix} 0 & 0 & 1 \\ -1 & -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

(b) Let this basis be $\{x, x_2, \dots, x_n\}$. Suppose $B = (e_1, e_2, \dots, e_n)$ is the standard basis of $\mathbb{R}^n$. Let

Remark 5.3.5 1. The projection map P_W depends on the complementary subspace W_0 .

Then prove that A is an orthogonal matrix. 8. Let $v, w \in \mathbb{R}^n$, $n \geq 1$ with $\|u\| = \|w\| = 1$. Prove that there

exists an orthogonal matrix A such that $Av = w$. Prove also that A can be chosen such that $\det(A) = 1$.

2. Observe that for a fixed subspace W , there are infinitely many choices for the complementary

5.3. ORTHOGONAL PROJECTIONS AND APPLICATIONS 101 3. It will be shown later that if V is an inner product space with inner product, $(\cdot, \cdot)$, then the subspace $\{(1, 1, 0, 1), (-1, 1, 1, -1), (0, 2, 1, 0), (1, 0, 0, 0)\}$ in $\mathbb{R}^4$.

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Theorem 5.3.10 Let S be a subset of a finite dimensional inner product space V , with inner product $(\cdot, \cdot)$.

2. Let S be equal to a subspace W . Then the subspaces W and $W^\perp$ are complementary. Moreover, if

$w \in W$ and $u \in W^\perp$, then $(u, w) = 0$ and $V = W + W^\perp$.

Let $\{w_1, w_2, \dots, w_k\}$ be a basis of W . By Gram-Schmidt orthogonalisation process, we get an orthonormal basis, say, $\{v_1, v_2, \dots, v_k\}$ of W . Then, for any $v \in V$,

2. $P_W[B, B]^2 = P_W[B, B]$, and

satisfies $P[B, B]^{\dagger} = P[B, B]$.

projection.